FALCON’s Response to Delhivery for RFQ of Bag Sorter_Bangalore

October 13, 2023

Kind Attention –

Mr. Sunny Raja

Mr. Himanshu Shekhar

M/s DELHIVERY Groupe

Offer Ref: F23-00450-01; Date 13.10.2023.

Subject – Techno -Commercial Offer of Bag Sorter

We are pleased to submit our Techno-Commercial Offer to DELHIVERY in response to your **RFQ for Bag Sorter, Bangalore.** We have taken this opportunity to also include adjustments discussed during our teams meeting on **05th Oct 2023**.

We would like to highlight the fact that Falcon Autotech, as a prime contractor for this project, is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative. From technical point of view, the system is **designed at 14000 BPH** and ensures simple operations and movement within the facility.

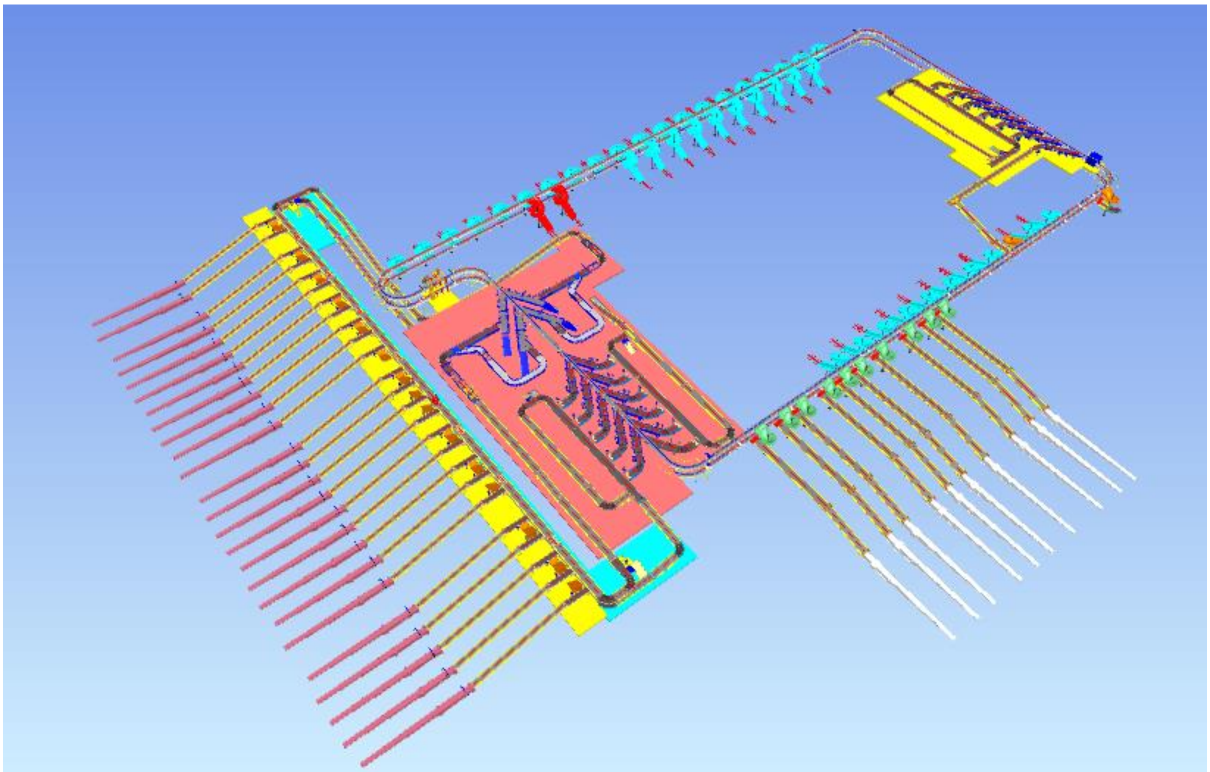
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Sandeep Bansal

Chief Business Officer

Response to RFQ for Bag Sorter_ Delhivery, Bangalore



Proposal Reference – F23-00450-01

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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Per Hour
3	ARB	Actuated Roller Balls
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Executive Summary

Falcon is pleased to confirm its great interest in responding to this RFQ of Bag Sorter Our team has been working closely with the relevant stakeholders, with a clear commitment to listening and understanding your needs and ensuring this project's success.
As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the Bag Sorter system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.
Our solution is based on the following key characteristics:

- **One loop of Bag sorter, based on a cross belt technology, offers a designed throughput of 14000 pph.**
- **Infeed conveyor system including 25 telescopic conveyors.**
- **58 gravity chutes, 70+ 50 (Optional) mini gravity chutes, 7 rejection chutes, 1 Strand Chute and 1 bulk chute.**

A tailor-made and simple layout, specifically designed to Delhivery The proposed layout, is the result of the technical requirements in the RFP document and our discussions with the relevant stakeholders during our last Teams workshop meeting on June 8th.

- Simple operational conditions due to one single loop cross belt sorter.
- Easy maintenance: optimized number of conveyors and concentrated inducts area.
- Vehicle movement zones.
- Exceptions handling area with direct feeding via conveyor.

1. **Falcon's reliable and fully CE compliant Parcel sortation systems**

These systems are globally being used by most innovative brands such as Aramex, Amazon, Asendia, Fastway, Delhivery and many more. The main and critical components of the FALCON Autotech sorter building blocks, like wheels, motors, belts, bearings, Bus Bars, Communication platforms, PLCs etc., are sourced from some of the best suppliers in the world, such as SEW, Siemens, SICK, Faigle, Vahle and Forbo. This strategic baseline of sourcing policy allows Falcon's customers to be fully confident in the systems' robustness and reliability.

2. **Commitment to quality systems**

Demonstrating Falcon's clear commitment to the Delhivery Group's satisfaction, the parcel sortation system, parts and services will be under warranty for 12 months from the completion of installation & commissioning.

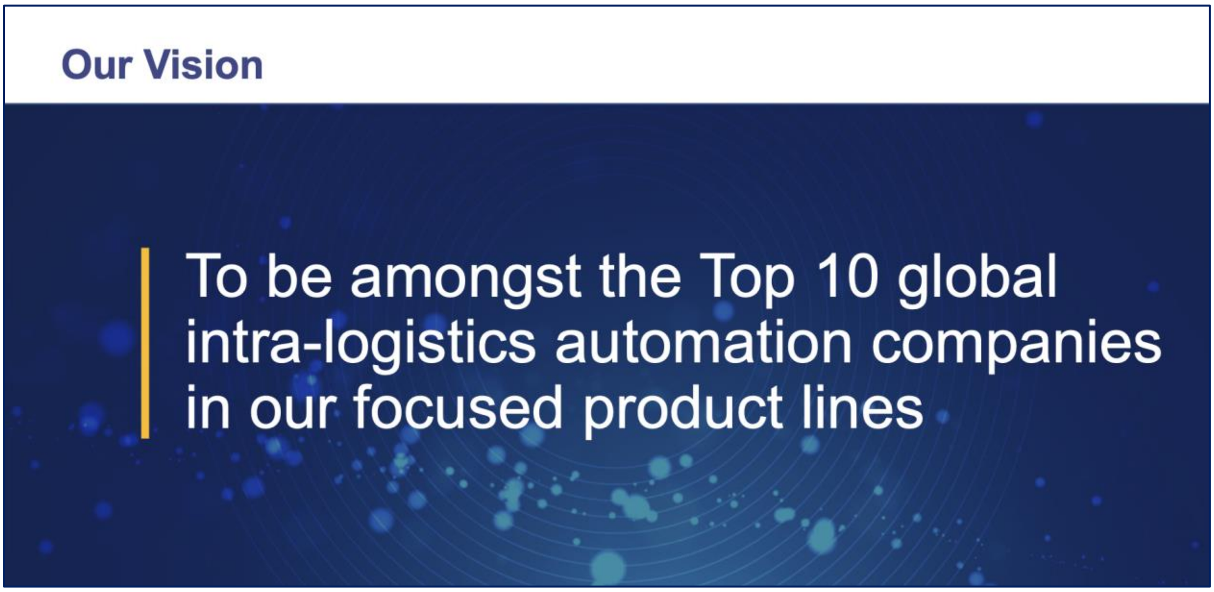
3. **Commitment to ensuring the quality of the delivery and its schedule.**

Falcon proposes a ramp-up approach, with a soft go-live by End of May 2025.

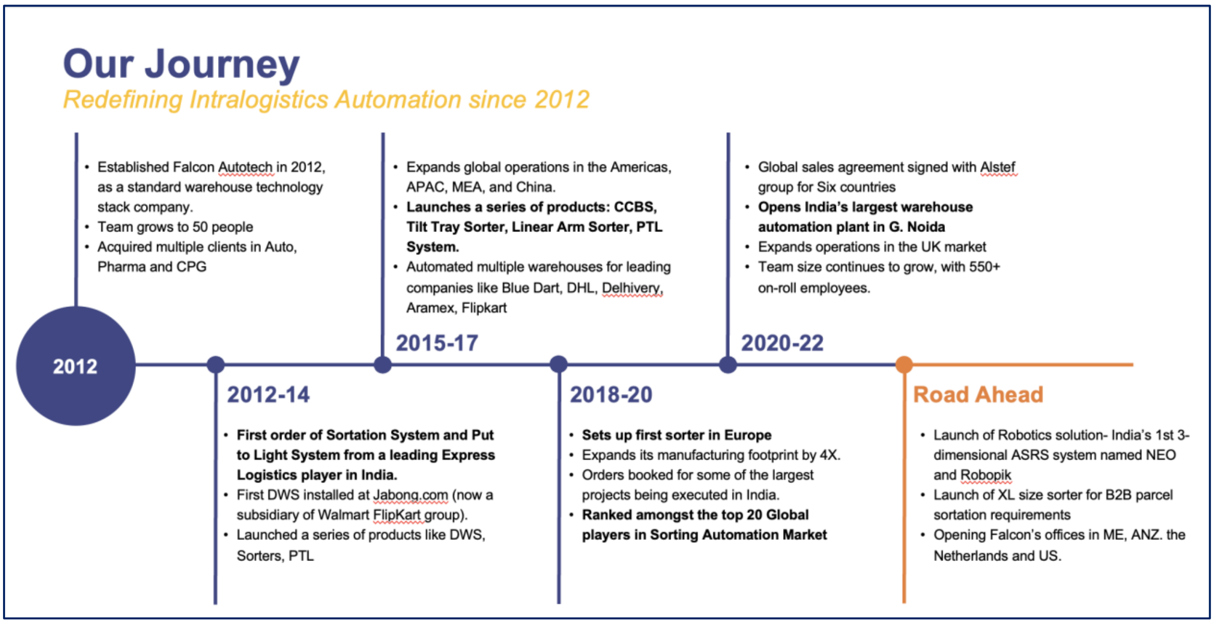
3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

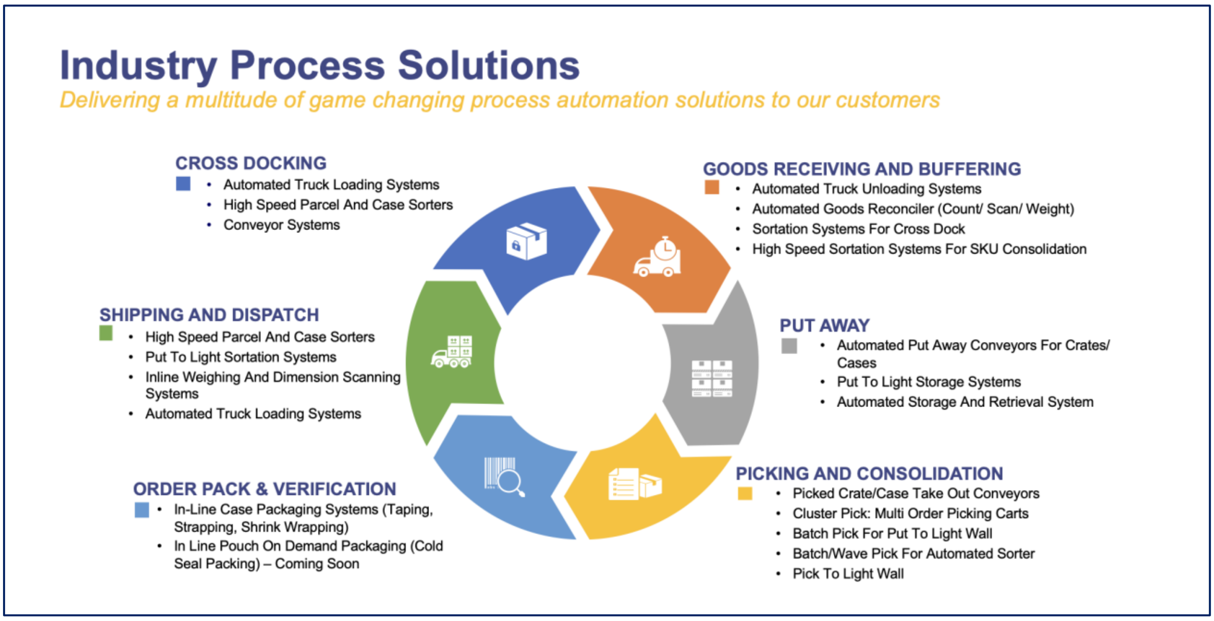
Falcon Autotech is currently among the top 15 intralogistics automation company, our vision is to become top 10 intralogistics automation company in our focused product lines.



The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. In order to be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.



Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of above product lines for effective materials handling, sortation and movement. The process is controlled in real-time by our in house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.

4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

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FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –

5.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –

5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –

5.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

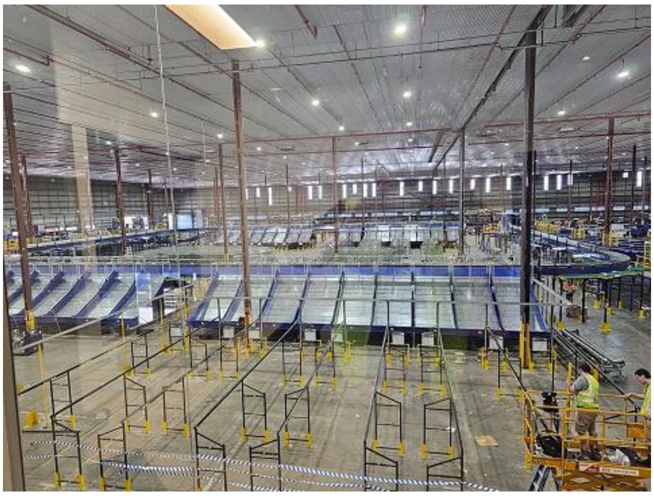
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –




5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

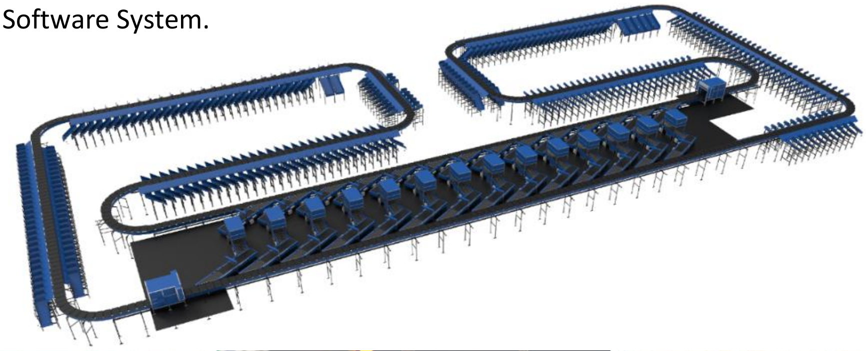
Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.

- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

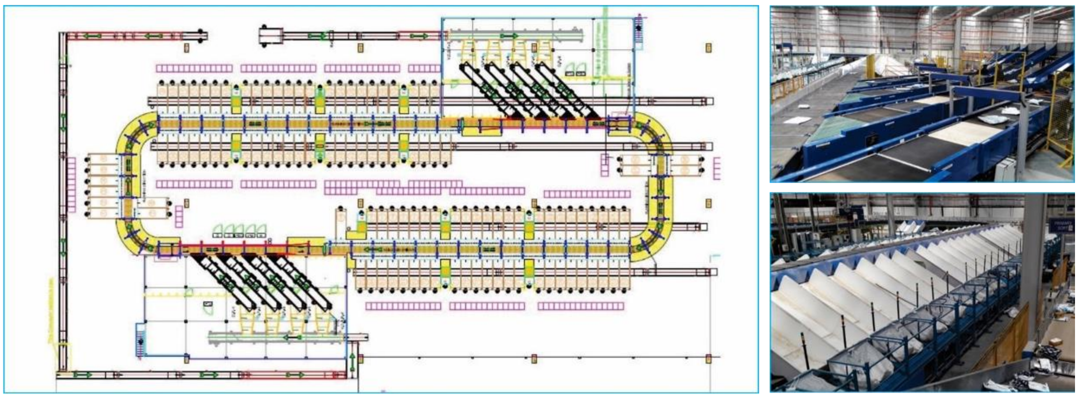
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Handled Parcel Spectrum

As per the parcel spectrum data provided in the RFQ documents, Falcon has studied and analysed the parcel spectrum in detailed.

Falcon purposes to use its "Heavy Dual Belt Cross Belt Sorter" to provide the maximum benefits to DELHIVERY Group in-terms of handling large parcel sizes and weight.

6.1 Parcel size loadable on the sorter

Falcon's heavy duty cross belt sorter has a capability to handle the below mentioned parcel sizes and weight.

Specification	Unit	Value
Max Length	mm	1000
Max Width	mm	800(850 for bags is allowed)
Max Height	mm	800
Max Weight	Kg	40
Min length	mm	150
Min Width	mm	150
Min Height	mm	50
Min Weight	Kg	0.2

Note- Dimension will only be captured for parcels > 20 mm in height .

6.2 Parcels to be loaded on Sorter shall have the following characteristics:

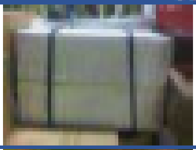
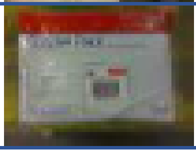
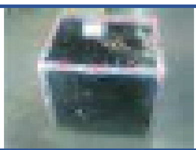
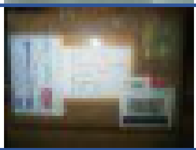
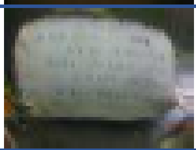
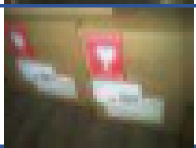
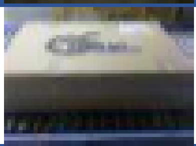
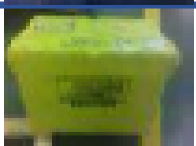
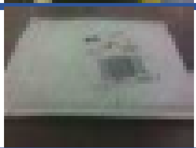
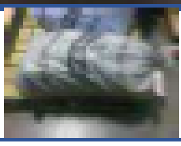
1. Centre of Gravity of item must not move during conveyance or sorting.
2. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
3. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
4. Plastic ropes shall be perfectly adherent to the surface of the package.
5. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
6. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
7. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
8. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
9. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
10. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

6.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

6.4 Pictures of Sortable Parcels

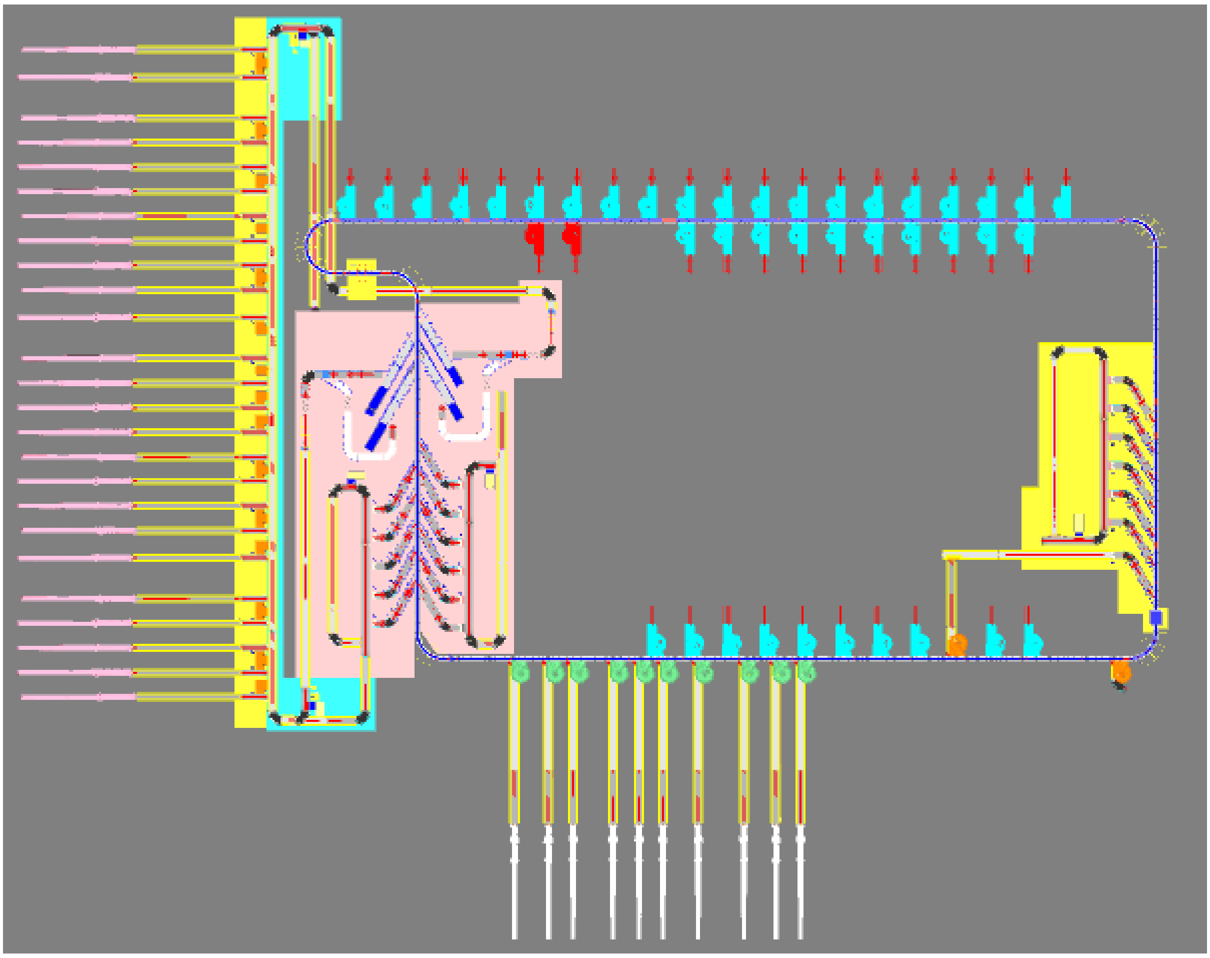
PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

7. Proposed System Description

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, and acceptance testing of the Loop CBS for sorting bags & boxes into destination as per specifications shared by DELHIVERY.

7.2 Summary of the System



The complete parcel processing system includes, in accordance with the specifications provided by **DELHIVERY:**

1. **Infeed System: -** Infeed System is divided in to two zones.
    - Zone 1 :
        - Bags & Boxes are loaded manually on the 25 telescopic belt conveyors, which are further connected to infeed highway conveyors.
        - These parcels travel from lower level to Mezzanine level and goes through aligning conveyors present before the auto-induct lines.
        - 25 manual segregation stations with sliding chutes are positioned strategically on the highway line where operators will segregate bags and

transfer them to the Bag Conveyor running below to the Box highway conveyor.
    - Boxes on the Common Highway Conveyor will move to the singulators where boxes coming in bulk will be singulated automatically.
    - Boxes will move to auto induct feedline (1 set on each deck) through which the boxes will induct on the sorter.
    - Bags are moved to dump VDS loop, from VDS loop operator will pickup the bags and scanned it manually on manual induct lines.

- Zone 2 :
    - Zone 2 constitutes of the outputs of the shipment sorter
    - Positive Bags prepared in the Shipment Sorter will come through the bag take Away conveyors of the shipment sorter and will be connected to VDS System
    - Large Size Parcels which may come at the inbound of the shipment sorters will be directly fed in the VDS by a separate conveyor, the input of which is below the VDS Loop Mezzanine
    - Operators at the induct loop will manually induct the bags and boxes to the CBS via the manual feedlines.

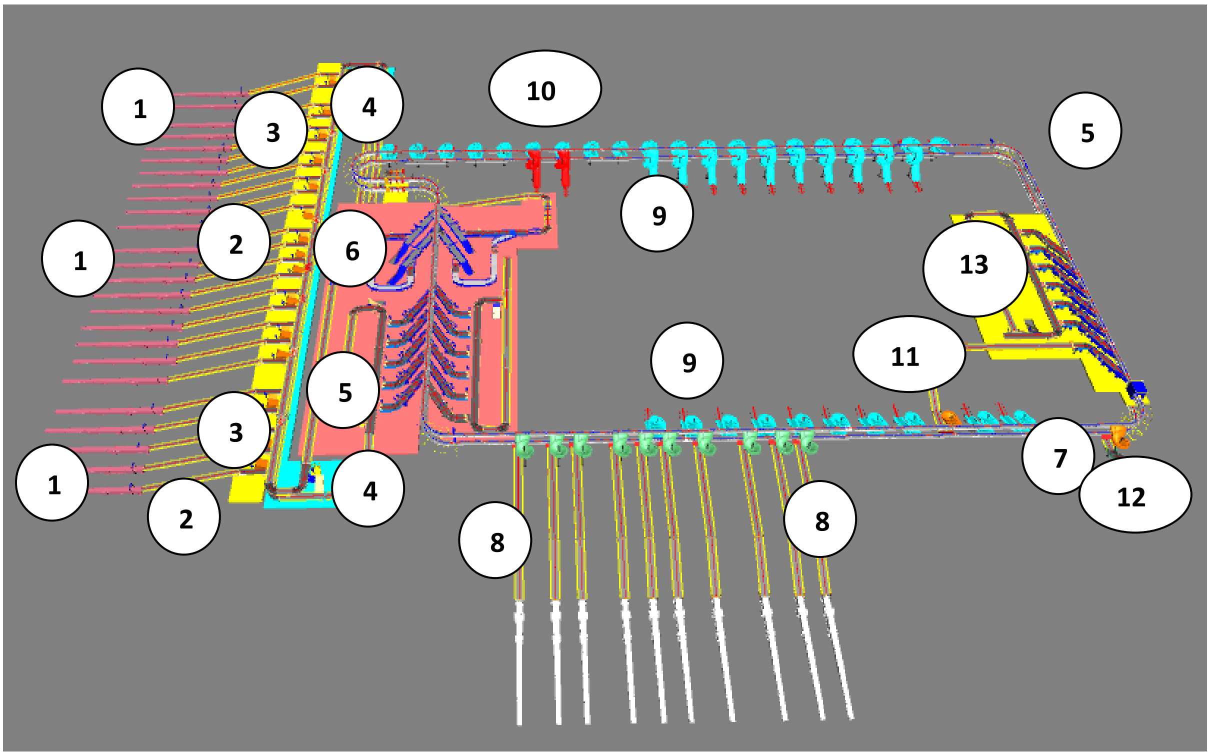
2. **Loop CBS:** - Once the parcels are on main loop, the CBS sorts the parcels into respective chutes using data from Delhivery Sorting logic.
3. **Sorting Output: -** The sorted shipments are discharged into three types of chutes.
    a. **Outbound Chute-** there are 10 Outbound chutes present in the system which will directly drops the bag 7 boxes on the outbound conveyor.
    b. **Staging Chute-** There are 40 Staging chutes in the system . The bags & boxes sorted on this chutes will collected and moved to the staging area manually .
    c. **Rejection Chute-** There are 2 Staging chutes type rejection chute in the system .
    d. **DBO Chute-** 1 DBO Chute is present to transfer the bags & boxes on the manual induct line of zone 2 for further process i.e. manual scanning.
    e. **Mixed Bag Chute-** One chute is planned to handle the bags having mixed parcels **.** This chute will drop the bags in to the conveyor which is connected to the infeed system of shipment sorter.

7.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.3 m/s.
3. Low occupancy of floor space in the building
4. Narrow discharge centers for increased number of splits in limited space
5. FALCON's CBS can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear.

8. Layout View



Legend

1 – Telescopic belt conveyor

2 – Infeed Inclined Conveyors

3 – Bag Segregation Station

4 – Infeed Highway Conveyor

5 – Upper Deck Zone 1 Manual Induct

6 – Upper Deck Zone 1 Auto Induct

7- Main Loop

8- Outbound Chute

9- Staging Chute

10- Rejection Chute

11- DBO Chute

12- Mixed Bag Chute

13- Upper Deck Zone 2 Manual Induct

9. Proposed System Capacity Calculations

9.1 Sorter System Capacity

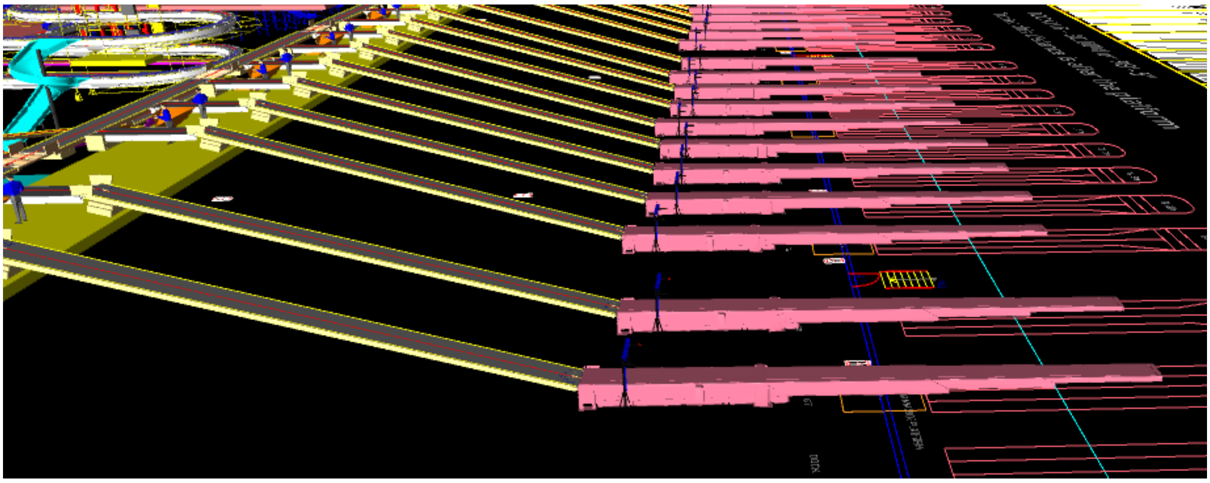
The following table shows the throughput calculation for the sortation system designed based on DELHIVERY's RFP requirements.

SPECIFICATION	VALUE
Sorter Type	**Loop CBS**
Speed	**2.3 m/s**
Pitch	**1,175 mm**
Carrier per hour	**7,047 CPH**
Belts per hour	**14,094 BPH**

10. System description

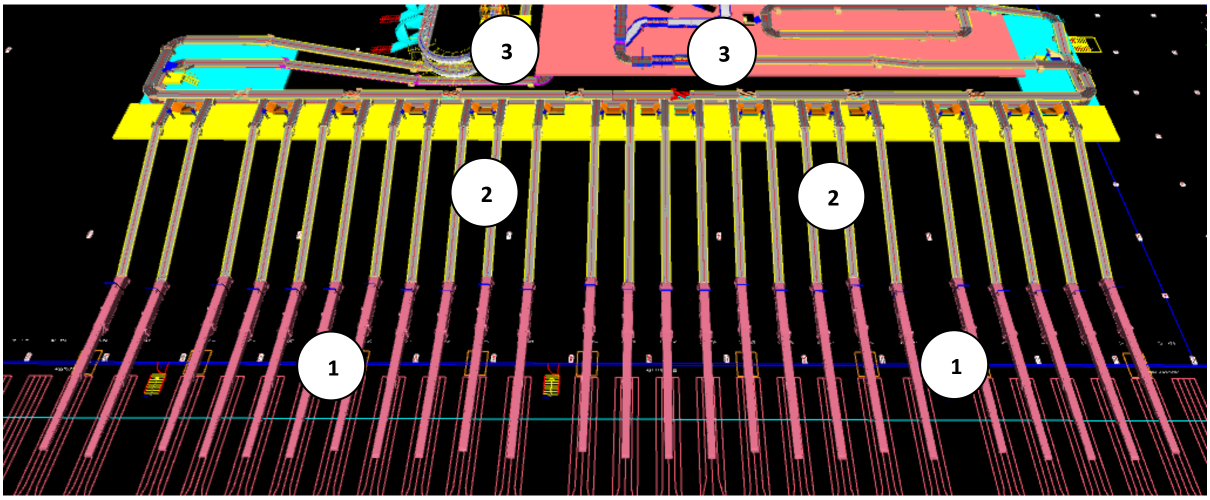
10.1 Unloading Area

Bags & Boxes arriving in Bulk in trucks will be unloaded onto one of the telescopic conveyors present on the dock. On each telescopic conveyor on sensor system is also planned to detect the oversize parcel parcels which will be removed by operator manually.



Truck Unloading Area

10.2 Infeed System



Legend

1- Telescopic Belt Conveyor
2- Infeed Inclined Conveyor system
3- Infeed Highway Conveyor

10.2.1 Telescopic Belt Conveyors

A telescopic belt conveyor, also known as an extendable conveyor or telescoping conveyor, is a type of conveyor system that features a telescoping mechanism to extend or retract the conveyor length as needed. It is commonly used in applications where flexible and adjustable conveying solutions are required, such as loading and unloading trucks or containers in warehouses, distribution centres, and shipping facilities.

In the proposed solution, Parcels from the Truck are transferred to infeed conveyors through Telescopic Belt Conveyors in singulated manner. TBCs are planned at a distance of 6 feet from dock leveller or 8 feet from the dock .

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech/Equivalent
Suitable Carton Size	MM	Min: 1000x800x800 Max: 100x100x5
Suitable Carton Weight	Kg	Min 0.05, Max 40
Base Length	Mm	7000
Extended Length (Total Length)	Mm	14000
Conveyor Belt Running Speed	m/min	18-30
Motor Rating	Type	0.75 Kw, 415V AC, 50 Hz
Load Capacity	Kg/m	40
Communication	Mm	2 x RS243, 1 x USB2.0, Ethernet

10.2.2 Infeed Conveyor System

Infeed conveyor system consists of below components:

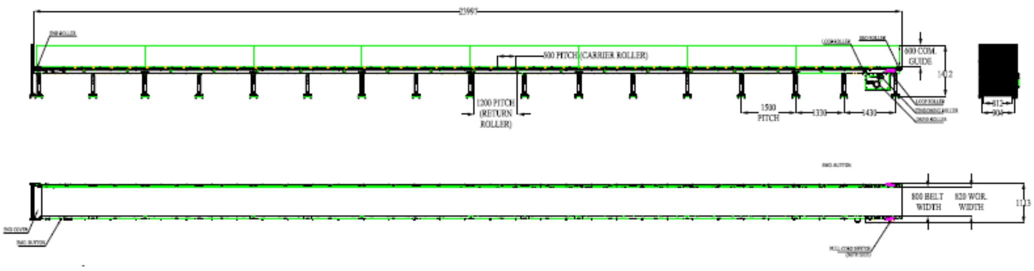
1. Inclined PVC Conveyor
2. Straight PVC Conveyor
3. Modular Conveyor
4. Curve Conveyor
5. Buffer Conveyor
6. Merge Conveyor

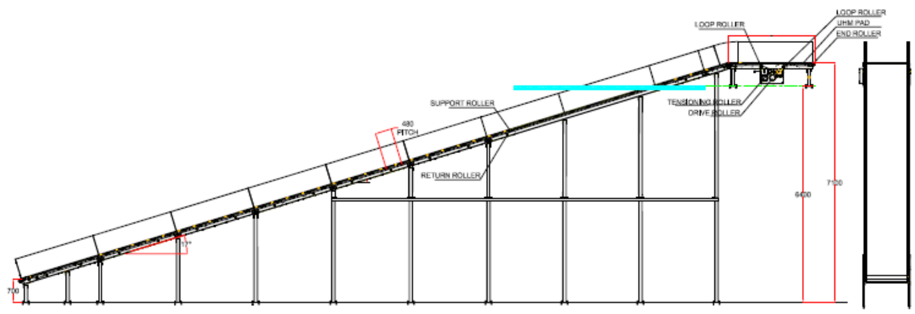
10.2.2.1 Straight and Inclined Powered Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.
Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.





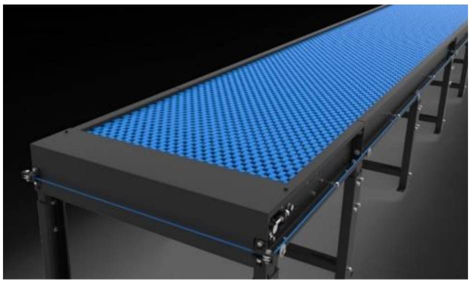
Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	1000/1200/1400
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns

Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Motor Shaft	MOC	EN9 Shaft material
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradly
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

10.2.2.2 Plastic Modular Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Modular Belt conveyor:



- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.

Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Plastic Modular Belt Material	Type	POM with PBT Pins, Pitch-25.4 mm, Pin Dia- 5mm
Plastic modular Belt Make	Make	Movex
Sprocket Size & No. of Sprockets on Drive End & Tail End	Type	5/146.4 mm
Sprocket material	MOC	POM
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns

Wear Strip Material	MOC	UHMW PE1000
Return Roller type	Type	DIA- 50 mm/20303
Return Roller Pitch	mm	800
Driven Shaft	Type	MOC-EN9 Solid Shaft hardened to BHN 240, Size-60SQ MM, Zinc Plated -25 microns, Bearings-UCF-210D1
Duplex Sprocket	MOC	MS
Duplex Chain	MOC	MS
Conveyor Under guarding	Type	GI SHEET (1MM THICK)
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.

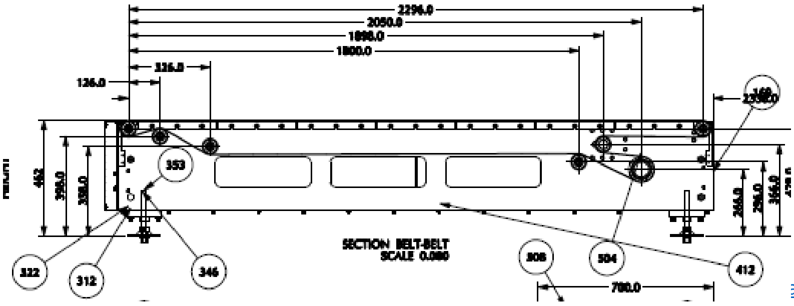
10.2.2.3 Curve Conveyor

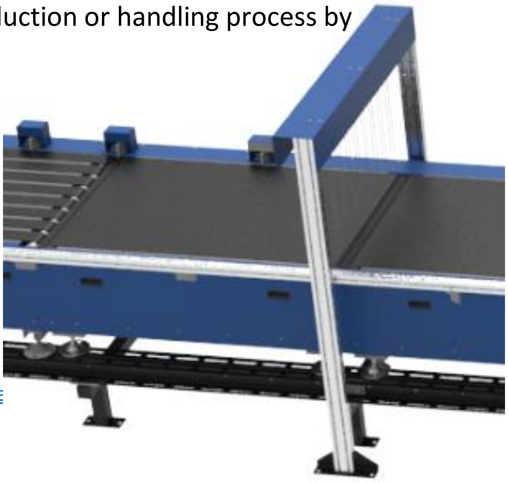
Falcon's curve conveyors are robust and easily maintainable. The uniquely designed curve and belts provides smooth environment to parcels for making the turns. The metal frames of the belts are not deformable to prevent belt misalignment. The belt guide assembly will include removable parts to allow quick replacement in case of damage.

10.2.2.4 Buffer Conveyor

Buffer conveyor is used to temporarily store or hold parcels in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

Buffer conveyor is particularly useful in scenarios where there is a discontinuity or variation in the flow rate of items. They allow for a smooth and continuous production or handling process by absorbing temporary surges or gaps in the flow.



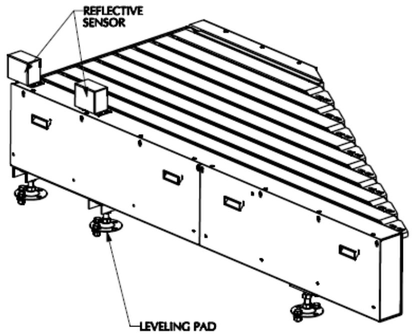


10.2.2.5 Belt Merge

This infeed belt conveyor can be used as a merging means with an angle of 30°, 45° and 60° in the same plane. It can be used, for example, as a follow-up conveyor to a sorter or as a feed unit in a collection circuit.

The nominal angle of this conveyor is achieved by belt modules of different lengths with self-centering belt belts. The optimum belt tension can be set individually for each belt band.Therefore, it is suitable for applications where a high transport speed is required.

The design of the guide rollers allows low-friction operation and low energy consumption; in addition, wear is reduced to a minimum.



10.2.2.6 Aligning Conveyor

Powered Aligning Conveyors are meant for aligning the parcels for ease and efficient induction on cross belt carrier. Aligning conveyors typically employ various mechanisms or techniques to achieve proper alignment.

By incorporating these alignment mechanisms, an aligning conveyor improves efficiency, reduces product damage, and enhances the overall accuracy of the production or handling process.

10.2.3 Conveyor BOQ

CONVEYOR BOQ PHASE -1							
S. NO.	NAME	ANGLE	EL_1	EL_2	Conveyor Length (m)	Conveyor Width (mm)	QTY.
1	Telescopic Belt Conveyor	A=0°			7+14		23
2	Idler Roller conveyor	A=0°	6281	6281	2.0		42
44	PVC Belt Conveyor	5.3	6605	6904	3.1	1410	1
45	PVC Belt Conveyor	0	4250	4250	18.2	1410	1
46	PVC Belt Conveyor	0	4250	4250	14.9	1410	1
47	PVC Belt Conveyor	0	4250	4250	7.7	1410	1
48	PVC Belt Conveyor	7.2	5792	6584	6.1	1410	1
49	PVC Belt Conveyor	0	5900	5900	13.4	1410	1
50	PVC Belt Conveyor	0	5900	5900	11.5	1410	1
51	PVC Belt Conveyor	0	5900	5900	6.2	1410	1
52	PVC Belt Conveyor	0	6600	6600	5.3	1410	1
53	PVC Belt Conveyor	1.9	6487	6902	12.3	1410	1
54	PVC Belt Conveyor	0	5900	5900	15.4	1410	1
55	PVC Belt Conveyor	0	4250	4250	7.7	1410	1
56	PVC Belt Conveyor	0	6500	6500	19.7	1410	1
57	PVC Belt Conveyor	0	3428	3428	4.3	1410	1
58	PVC Belt Conveyor	16	770	6050	19.0	1010	1
59	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
60	PVC Belt Conveyor	16	770	6050	19.0	1010	1
61	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
62	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
63	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
64	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
65	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
66	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
67	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
68	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
69	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
70	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
71	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
72	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
73	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
74	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
75	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
76	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
77	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
78	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1

79	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
80	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
81	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
82	PVC Belt Conveyor	-13.8	3400	1047	9.7	1410	1
83	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
84	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
85	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
86	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
87	PVC Belt Conveyor	16	770	6050	19.0	1010	1
88	PVC Belt Conveyor	16	770	6050	19.0	1010	1
89	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
90	PVC Belt Conveyor	0	3500	3500	18.2	1410	1
91	PVC Belt Conveyor	16.2	3328	6735	12.0	1410	1
92	PVC Belt Conveyor	0	3428	3428	2.4	1410	1
93	PVC Belt Conveyor	0	6500	6500	9.6	1410	1
94	PVC Belt Conveyor	12.5	4147	5337	5.3	1410	1
95	PVC Belt Conveyor	0	6500	6500	19.7	1410	1
96	PVC Belt Conveyor	0	5900	5900	2.4	1410	1
97	PVC Belt Conveyor	0	6600	6600	18.2	1410	1
98	Modular Belt Conveyor	0	6600	6600	9.2	1410	1
99	PVC Belt Conveyor	0	5900	5900	7.7	1410	1
100	PVC Belt Conveyor	0	5900	5900	12.0	1410	1
101	PVC Belt Conveyor	8.5	5263	6684	9.5	1410	1
102	PVC Belt Conveyor	0	4250	4250	19.7	1410	1
103	PVC Belt Conveyor	0	6600	6600	3.8	1410	1
104	PVC Belt Conveyor	0	6600	6600	19.7	1410	1
105	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
106	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
107	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
108	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
109	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
110	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
111	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
112	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
113	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
114	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
115	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
116	Belt Turn(90 deg)	A=0°	6600	6600		1400	11
118	Belt Turn(60 deg)	A=0°	6500	6500		1200	12
121	Modular Belt Conveyor	0	6600	6600	8.3	1400	1
122	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
123	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
124	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
125	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
126	Modular Belt Conveyor	0	5970	5970	4.6	1010	1

127	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
130	Aligning Conveyor		5900	5900	2.0	1400	4
133	Modular Belt Conveyor	0	6600	6600	10.4	1400	1
134	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
135	Modular Belt Conveyor	0	5900	5900	4.0	1400	1
138	Modular Belt Conveyor	0	6600	6600	2.5	1400	1
139	Modular Belt Conveyor	0	6600	6600	19.6	1400	1
144	Belt Turn(90 deg)	A=0°	6500	6500		1200	3
146	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
147	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
148	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
149	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
150	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
151	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
152	Modular Belt Conveyor	0	6600	6600	19.6	1400	1
154	Singulator		6500	6500			1
155	Aligning Conveyor		6500	6500	2.7	1200	2
157	Spacing Conveyor		6500	6500		1200.0	4
159	Buffer Conveyor		6500	6500		1200.0	4
164	PVC Belt Conveyor		6500	6500	1.4	1200.0	1
166	PVC Belt Conveyor		6500	6500	4.5	1200.0	1
168	PVC Belt Conveyor		6500	6500	3.2	1200.0	1
169	PVC Belt Conveyor		6500	6500	3.2	1200.0	1
173	Belt Merge		6500	6500		1200.0	1
175	Break roller						84

CONVEYOR BOQ PHASE 2							
S. NO.	NAME	ANGLE	EL_1	EL_2	Conveyor Length (m)	Conveyor Width (mm)	QTY.
1	Telescopic Belt Conveyor	A=0°					12
2	PVC Belt Conveyor	0	4000	4000	1.4	1410	1
3	PVC Belt Conveyor	10.6	894	4083	17.2	1410	1
4	PVC Belt Conveyor	0	4000	4000	2.4	1410	1
5	PVC Belt Conveyor	0	3850	3850	19.7	1410	1
6	PVC Belt Conveyor	0	3850	3850	12.5	1410	1
7	PVC Belt Conveyor	0	4250	4250	16.8	1410	1
8	PVC Belt Conveyor	0	4250	4250	19.7	1410	1
9	PVC Belt Conveyor	-10	4153	1112	17.3	1410	1
10	PVC Belt Conveyor	0	1000	1000	19.7	1410	1
11	PVC Belt Conveyor	0	1000	1000	8.6	1410	1
12	PVC Belt Conveyor	-13.8	5800	1109	19.6	1410	1
13	PVC Belt Conveyor	0	5900	5900	3.8	1410	1

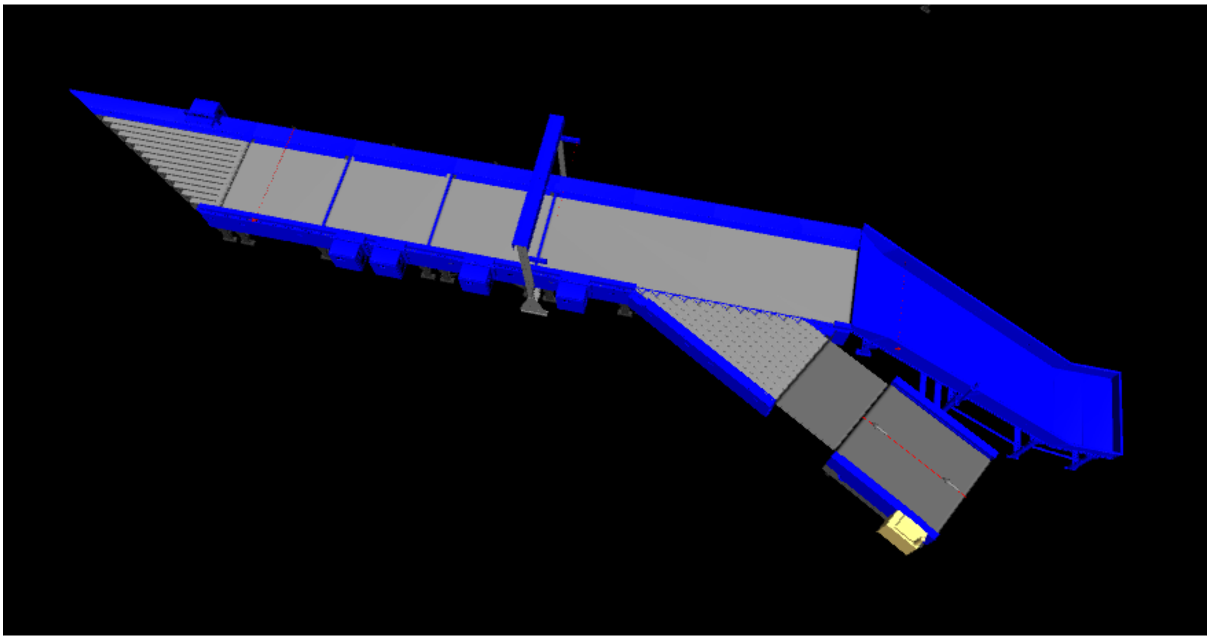
14	PVC Belt Conveyor	0	5900	5900	13.4	1410	1
15	PVC Belt Conveyor	0	5900	5900	10.6	1410	1
16	PVC Belt Conveyor	0	5900	5900	12.0	1410	1
17	PVC Belt Conveyor	0	5900	5900	9.6	1410	1
18	PVC Belt Conveyor	0	1000	1000	9.6	1410	1
19	PVC Belt Conveyor	16	770	6050	19.0	1010	1
20	PVC Belt Conveyor	16	770	6050	19.0	1010	1
21	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
22	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
23	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
24	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
25	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
26	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
27	PVC Belt Conveyor	15.5	769	6050	19.7	1010	1
28	PVC Belt Conveyor	16	770	6050	19.0	1010	1
29	PVC Belt Conveyor	16	770	6050	19.0	1010	1
30	PVC Belt Conveyor	16	770	6050	19.0	1010	1
31	PVC Belt Conveyor	0	4250	4250	4.0	1410	1
32	PVC Belt Conveyor	0	1000	1000	12.0	1410	1
33	PVC Belt Conveyor	0	5900	5900	11.5	1410	1
34	PVC Belt Conveyor	13.3	897	3931	13.0	1410	1
35	PVC Belt Conveyor	0	1000	1000	16.3	1410	1
36	PVC Belt Conveyor	0	4250	4250	13.4	1410	1
37	PVC Belt Conveyor	0	4250	4250	19.7	1410	1
38	PVC Belt Conveyor	7.4	3857	4184	2.4	1410	1
39	PVC Belt Conveyor	0	1000	1000	15.8	1410	1
40	Belt Turn(90 deg)					1400	11
41	Modular Belt Conveyor	0	4100	4100	19.6	1400	1
42	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
43	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
44	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
45	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
46	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
47	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
51	Modular Belt Conveyor	0	4100	4100	3.2	1400	1
52	Modular Belt Conveyor	0	4100	4100	8.4	1400	1
53	Modular Belt Conveyor	0	5900	5900	6.8	1400	1
55	Aligning Conveyor		5900	5900	2.0	1400	4
62	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
63	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
64	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
65	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
66	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
67	Modular Belt Conveyor	0	5970	5970	4.6	1010	1
72	PVC Belt Conveyor	0	6500	6500	2.2	1200	1

73	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
74	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
75	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
76	PVC Belt Conveyor	0	6500	6500	2.2	1200	1
77	Belt Turn(60 deg)	A=0°	6500	6500		1200	5
78	Singulator		4000	4000			1
81	Spacing Conveyor		4000	4000		1200.0	4
83	Buffer Conveyor		4000	4000		1200.0	3
87	PVC Belt Conveyor		4000	4000	1.4	1200	1
88	Belt Turn(90 deg)		4000	4000		1200	3
89	PVC Belt Conveyor		4000	4000	4.5	1200	1
91	PVC Belt Conveyor		4000	4000	3.2	1200	1
92	PVC Belt Conveyor		4000	4000	3.2	1200	1
93	Aligning Conveyor		4000	4000	2.7	1200.0	2
94	Belt Turn(60 deg)		4000	4000		1200	1
97	Belt Merge		4000	4000		1200.0	1

10.3 Feedlines

The System has a two type of Feedlines :

- **Auto Induct Feedlines:**



1. Each Auto Induct has Eight Conveyor Modules-

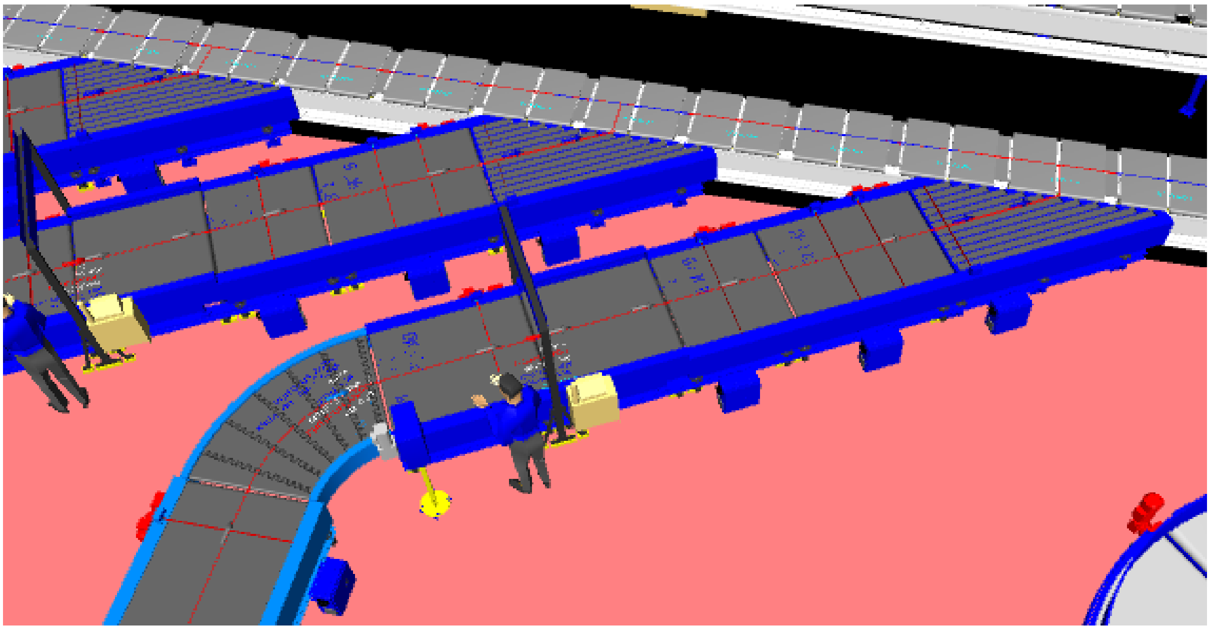
   a) Spacer Conveyors- 2 Nos.
   b) Short Buffer conveyor- 2 Nos.
   c) Weighing Conveyor- 1 No.
   d) Receiving Conveyor- 1 No.
   e) Angle merge conveyor- 2 No.

2. Rejection chute is provided to take out oversized and overweight parcels from the system.

3. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.

4. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop.

- **Manual Induct Feedlines:**



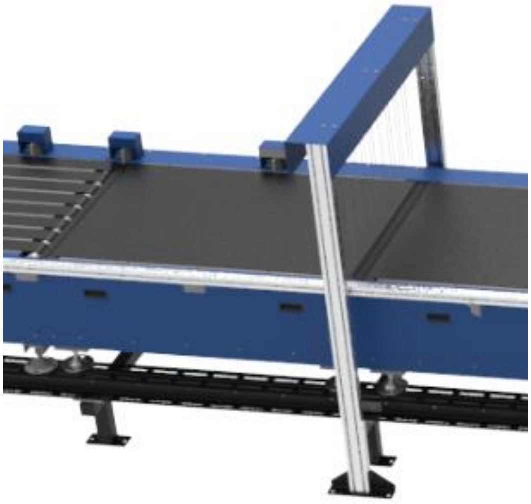
Each Manual Induct has 5 Conveyor Modules-

f) Loading Conveyor – 1 No
g) Weighing Conveyor- 1 No
h) Spacer Conveyors- 2 Nos.
i) Angle merge conveyor- 1 No.

10.3.1 Spacing Conveyor

A spacing conveyor, also referred to as a gapping conveyor or gap optimizer, is a type of conveyor system used to create and maintain consistent gaps or spacing between items as they move along the conveyor line. Its primary purpose is to regulate the flow and spacing of products to ensure smooth operation and efficient downstream processes.

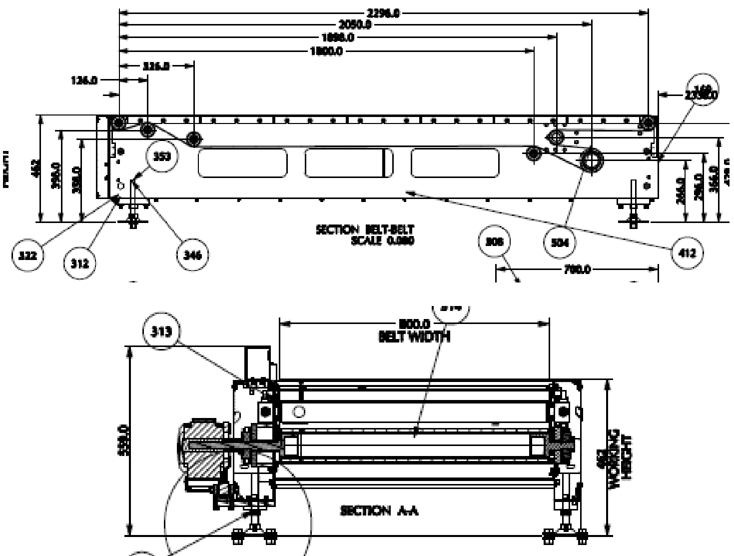
This conveyor is a variable speed special purpose module that creates space between parcels as well as regulates feeding on the Weighing conveyor.



10.3.2 Buffer Conveyors

A buffer conveyor, also known as a buffering conveyor or accumulation conveyor, is a type of conveyor system used to temporarily store or hold items in a controlled manner. Its primary purpose is to manage the flow of items between different stages of a production or handling process when there is a mismatch in the speeds or capacities of the upstream and downstream equipment.

This Conveyors required to maintain the Throughput of Line.



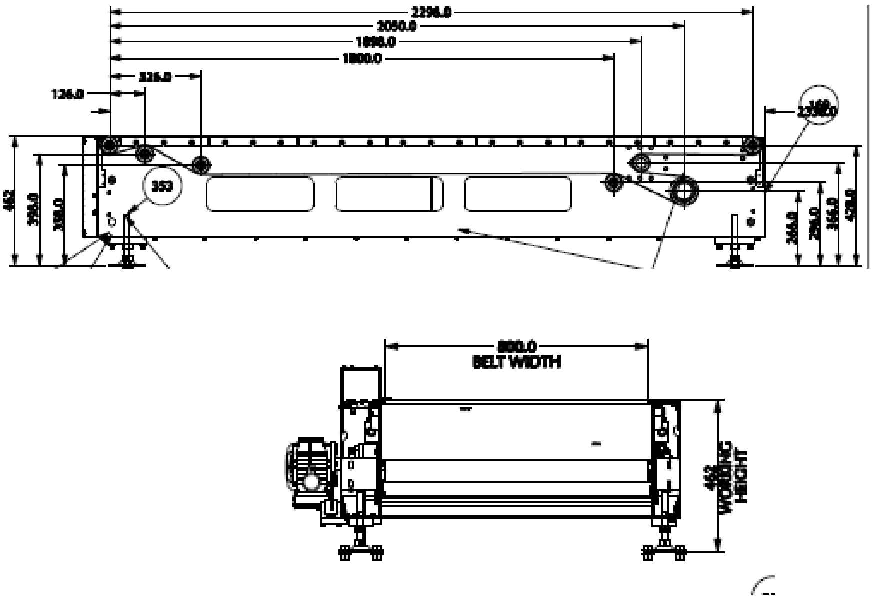
A weighing conveyor, also known as a weigh belt conveyor, is a type of conveyor system specifically designed to measure the weight of materials as they move along the conveyor belt. It combines the functions of conveying and weighing into a single integrated process.

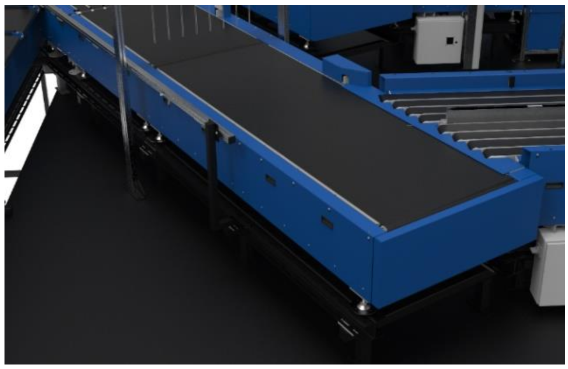
Weighing Conveyors equipped with high precision Load Cells to capture the weight of Parcels.

Make- Bizerba/ Equivalent

10.3.4 Receiving Conveyor

A receiving conveyor, also known as an inlet conveyor or an intake conveyor, is a type of conveyor system used to receive and transfer products from the infeed-line into a processing line. It serves as the initial point of entry where products are collected and conveyed to subsequent stages of the process.

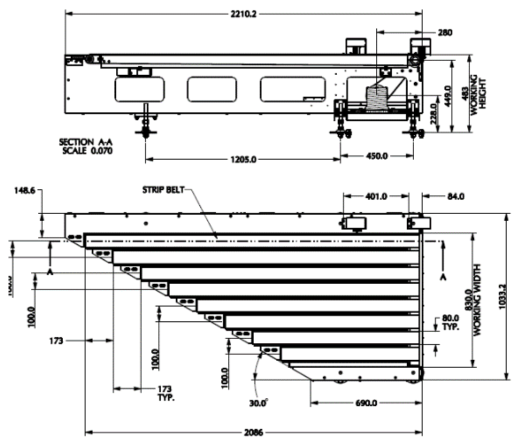




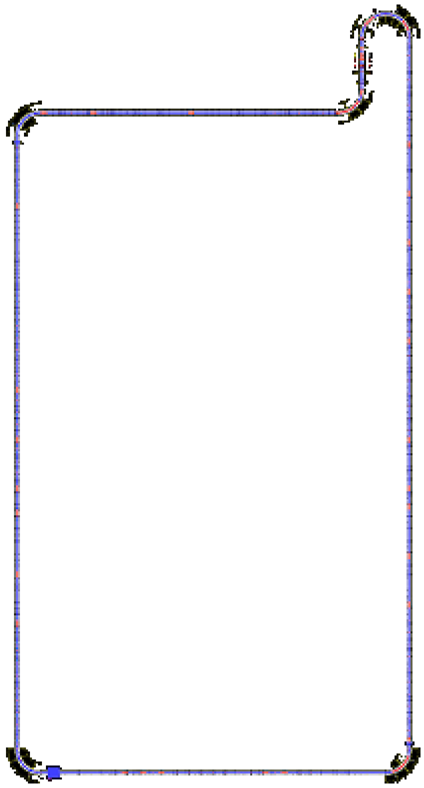
10.3.5 Intelligent Merge Conveyor

An intelligent merge conveyor is a type of conveyor system that incorporates advanced automation and control technologies to intelligently merge multiple conveyor lines or streams of materials into a single unified flow. It optimizes the merging process by dynamically adjusting the speed and position of items to ensure a smooth and efficient merge.

In the proposed solution this is 30° triangular Dual Belt high-speed conveyor used for inducting parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.



10.4 Main Loop



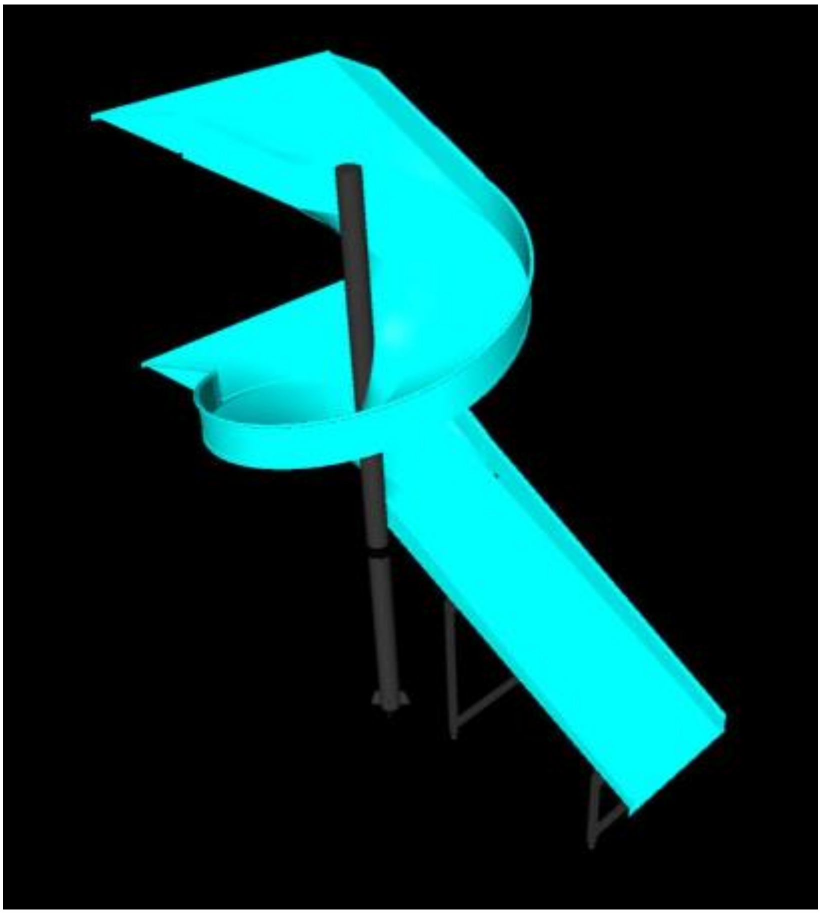
1. The proposed Single Deck CBS Sorter will be installed on ground level. The upper deck Sorter runs at 6500 mm from ground level and lower deck on 4000 mm from ground level.

2. We have provided loop with dual Belt Carriers with 1175 mm pitch and Belt Size of 495 x 800mm.

3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

10.5 Sorting Output

10.5.1 Staging Chute



- THESE ARE COLLECTION TYPE CHUTES WHERE THE BAGS/BOXES KEEP COLLECTING INTO IDLER ROLLER CONVEYOR AS THE SORTING CONTINUES.
- THE USER COLLECTS THE BOXES/BAG FROM THE ROLLER CONVEYORS.

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Three Color Tower Light

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status

10.5.2 Outbound Chute

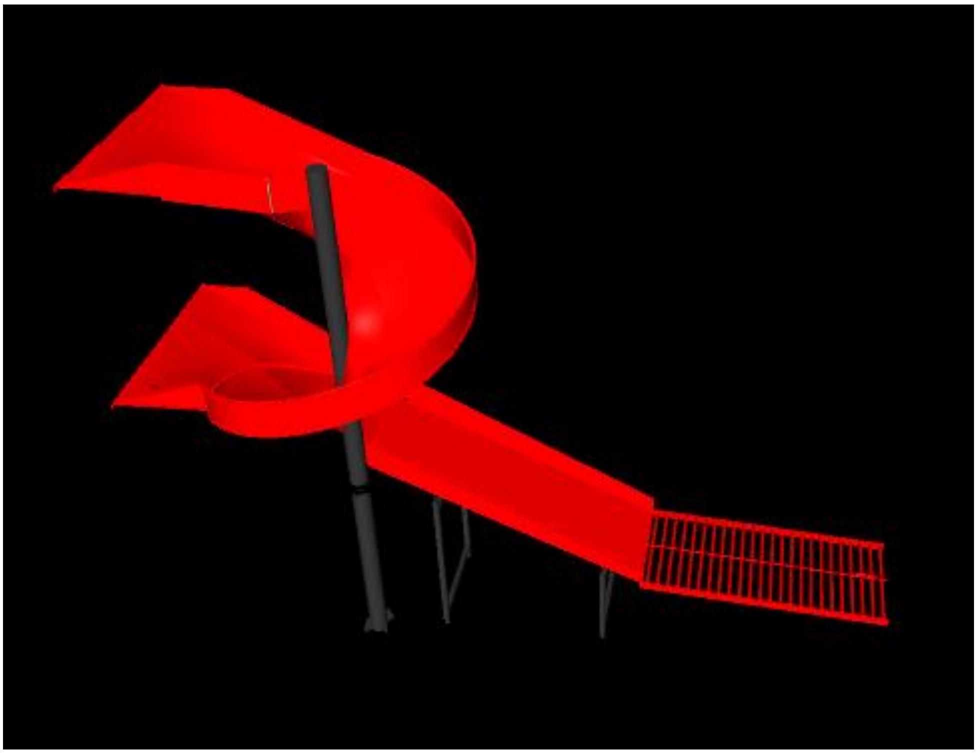
- These are direct takeout chutes where the bags/boxes will be drop directly on the takeout conveyors.
- Takeout conveyors will drop the bags/boxes on the telescopic conveyor in turn which are connected to the docks for loading into trucks

Each Chute is equipped with below components to ease the operations:

- Chute Full Sensor
- Three Color Tower Light

Chute I/Os	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation
Three Colour Tower Lamp	To indicate the chute status

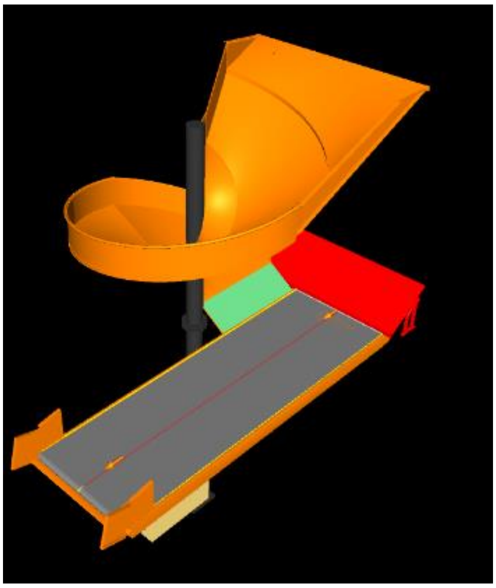
10.5.3 Rejection Chute



System is installed with 2 Nos rejection chutes (staging type chute). Once the parcel got rejected, it will stage on the roller conveyors for the manual process.

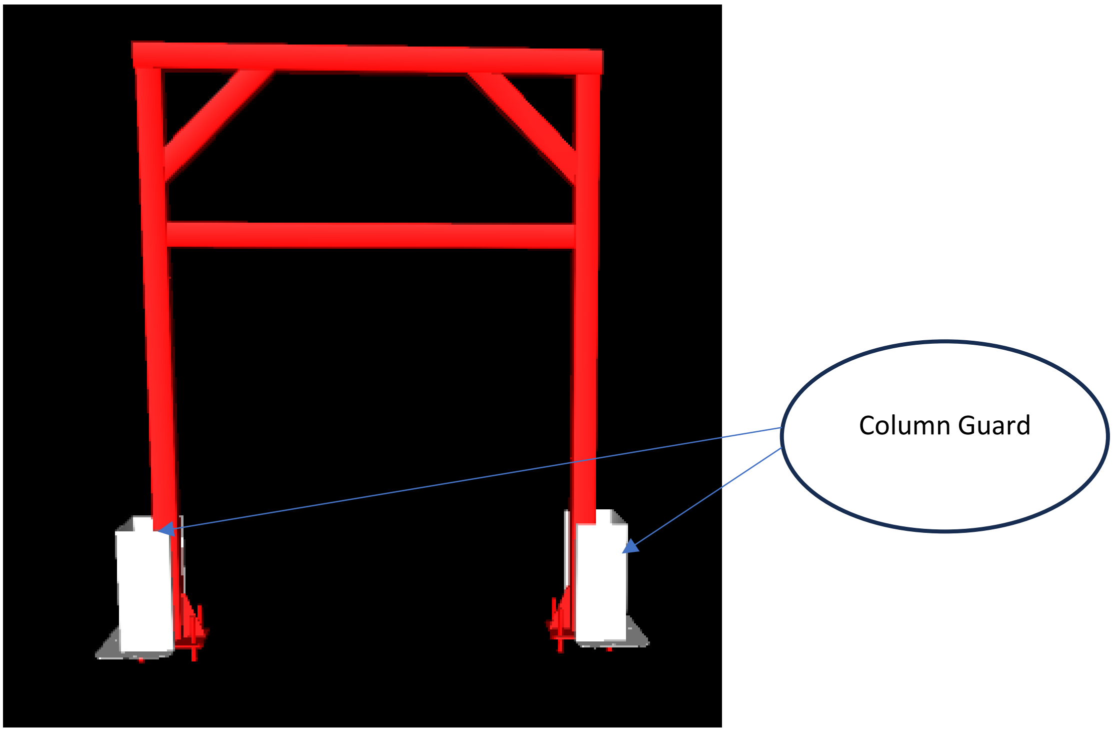
10.5.4 DBO Chute

The system is designed with 1 DBO chute. As per the requirement of DELHIVERY, DBO parcels will transfer to the dedicated manual induct for refeeding through after manual scanning.



10.6 Protection Against Collision):

In our solution, to avoid the collision or for the protection of system against the trolley or vehicle collision we have considered column guards on each leg where the vehicle and trolley movement are planned.



Sample Image for Reference Only

11. Proposed System Technical Details

11.1 Mechanical equipment

Phase 1			
Pos.	**Qty.**	**Description**	**Value**
FM1	**1**	**Infeed System** • Powered Belt Conveyors • Modular Belt Conveyor • Merge Conveyor • Aligning Conveyor • Curve Conveyor- 90 Degree • Curve Conveyor- 60 Degree • Buffer Conveyor • Spacing Conveyor • Idler Roller Conveyor • Singulator • Break Roller • Telescopic Belt Conveyor (7+14)	~857 Metres (75 Conveyors) ~138 Metres (20 Conveyors) 1 No 6 Nos 14 Nos 12 Nos 4 Nos 4 Nos 84 metres (42 Nos) 1 No 84 Nos 23 Nos .
FM2	**11**	**Manual Induct Feedlines** Consists of • Loading Conveyor. • Weighing System • Dynamic Spacing System • Angle merge • 2-D Hand Scanners for Manual Scanning	1 Set 1 Set 2 Set 1 Set 1 No.
FM3	**2**	**Auto Induct Feedlines** Consists of • Weighing Conveyor • Spacing System • Receiving Conveyor • Buffer Conveyor • Angle Merge	1 No. 2 Set. 1 Nos 2 Nos 2 Nos 1 No
FM4	**1**	**Falcon Upper Deck Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch Including:	6500 mm ~438 m ~2.3 m/s Linear Motor 1175 mm

		• Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Camera based Barcode reader. • Dimension & 5 side Scanning System • E-Stops • Hooters • Fencing • Netting • Overshoot Assembly	
FM5	**1**	**Falcon Lower Deck Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length Including: • Standard Sorter Supports	 4000 mm ~438 m
FM6	**1**	**Sorter Outputs** • Outbound Chutes • Staging Chute • Mixed Bag Chute • Rejection Chutes • DBO Chute • Rejection Chute – Auto Induct • Rejection Chute- Manual Induct • Sliding Chute (Bag Segregation on Infeed System) • Chute Full sensors • Tower Lamp (3 Colour)	 10 Nos 40 Nos 1 No 2 Nos 1 No 2 Nos 2 Nos 10 Nos

Phase 2			
Pos.	**Qty.**	**Description**	**Value**
FM1	**1**	**Infeed System** • Powered Belt Conveyors • Modular Belt Conveyor • Merge Conveyor • Aligning Conveyor • Curve Conveyor- 90 Degree • Curve Conveyor- 60 Degree • Buffer Conveyor • Spacing Conveyor • Singulator • Telescopic Belt Conveyor (7+14)	 ~577 Metres (47 Conveyors) ~94 Metres (16 Conveyors) 1 No 6 Nos 14 Nos 6 Nos 3 Nos 4 Nos 1 No 10 Nos .

FM2	**5**	**Manual Induct Feedlines** Consists of • Loading Conveyor. • Weighing System • Dynamic Spacing System • Angle merge • 2-D Hand Scanners for Manual Scanning	1 Set 1 Set 2 Set 1 Set 1 No.
FM3	**2**	**Auto Induct Feedlines** Consists of • Weighing Conveyor • Spacing System • Receiving Conveyor • Buffer Conveyor • Angle Merge	1 No. 2 Set. 1 Nos 2 Nos 2 Nos 1 No
FM4	**1**	**Falcon Lower Deck Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch Including: • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Camera based Barcode reader. • Dimension & 5 side Scanning System • E-Stops • Hooters • Fencing • Netting • Overshoot Assembly	4000 mm ~438 m ~2.3 m/s Linear Motor 1175 mm
FM6	**1**	**Chutes** • Rejection Chute – Auto Induct • Rejection Chute- Manual Induct • Sliding Chute (Bag Segregation on Infeed System)	2 Nos 1 No 7 Nos

11.2 Electrical Equipment

Electricals			
FE1	**1**	**Consists of** Main Power Distribution panel Main Control Panel Feedline Control Panels Sorter Drive Panels Network Switches Field Cabling	Included

11.3 Control System

Components			
FC1	**1**	**Consists of** Siemen's PLC based Control System with SCADA Industrial Switch	1 Nos. As per requirement

12. Description of Components of Equipment

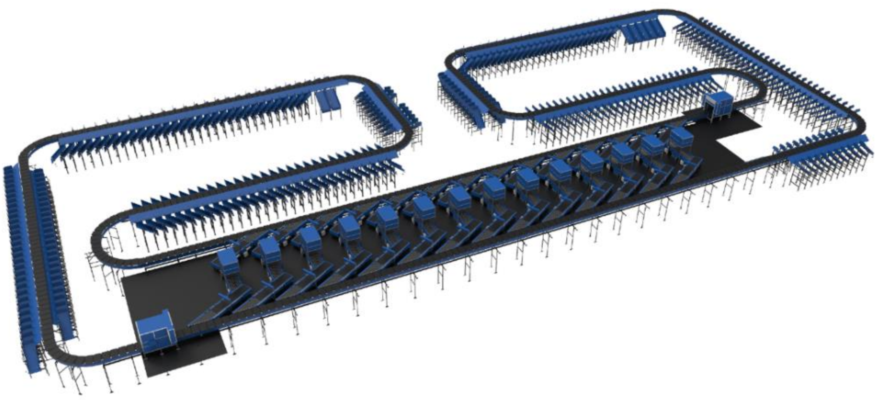
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

12.1 Cross Belt Sorter -

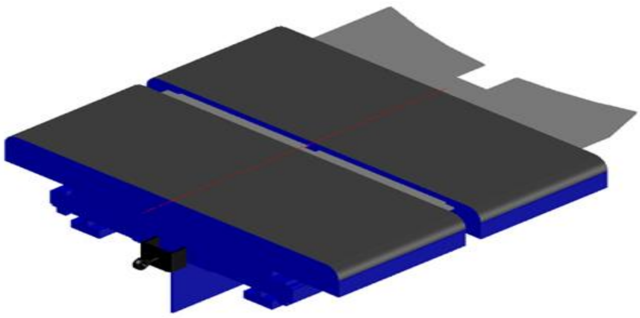
Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels. Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements



12.2 CBS Carrier

Falcon Autotech's CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable parcel recirculation.



12.2.1 Servo Roller

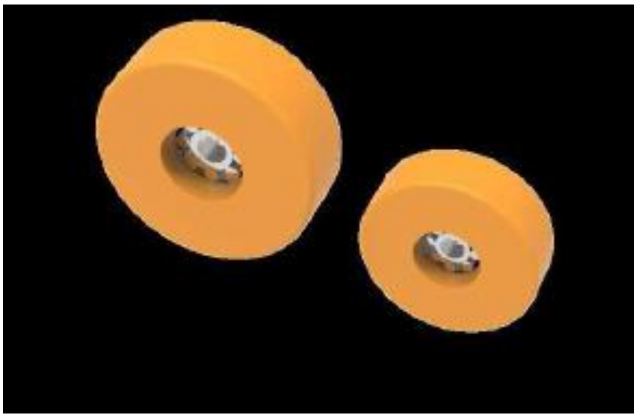
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

12.2.2 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

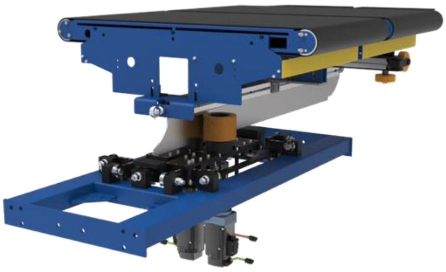
12.2.3 Carrier Wheels

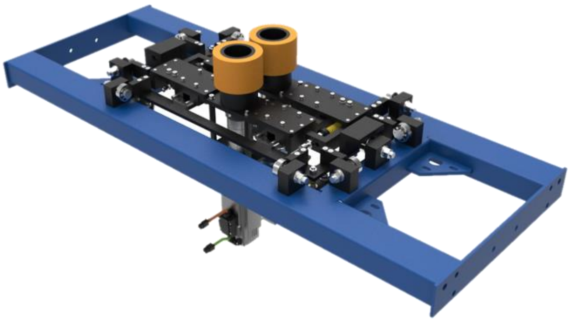
Carrier wheels that are thoroughly tested and proved for long life cycle.



12.2.4 Friction Wheel Drive

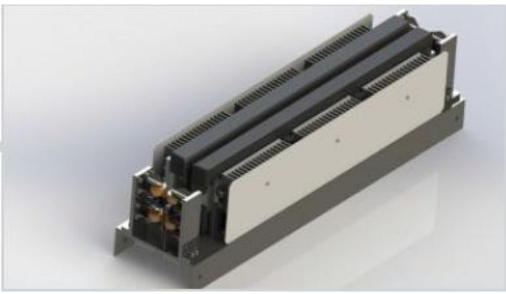
Falcon can provide the indigenously developed Friction Wheel Drive (FWD) to drive the cross belt loop. This driving mechanism operates on the principle of friction, as the name implies. This unit comprises two independent motor-driven wheels that spin in opposite directions synchronised. FWDs are highly energy-efficient drives that promote **sustainability** compared to Linear induction/ synchronous motors.





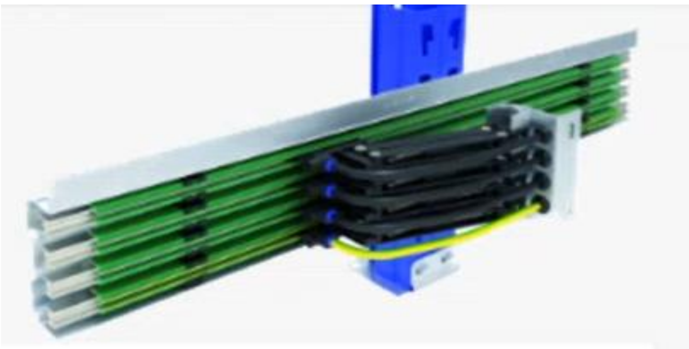
12.2.5 Non-contact based linear motor drive

No Contact based Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed.



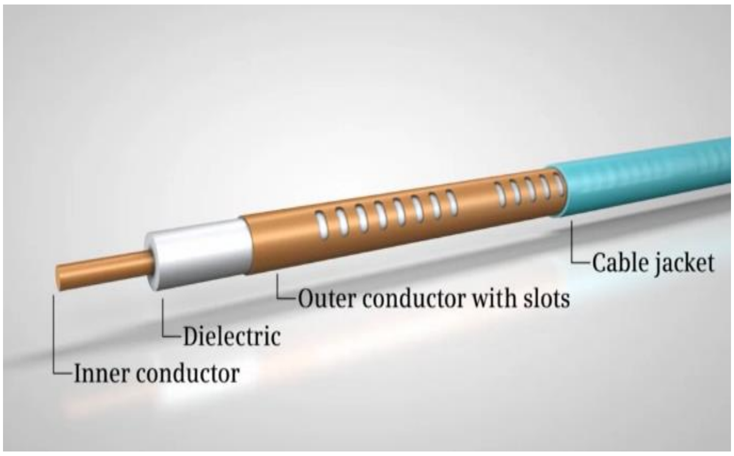
12.2.6 Power Transmission

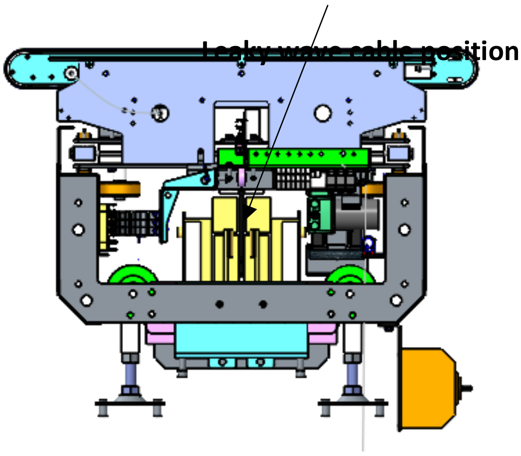
Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.



12.2.7 Data Transmission

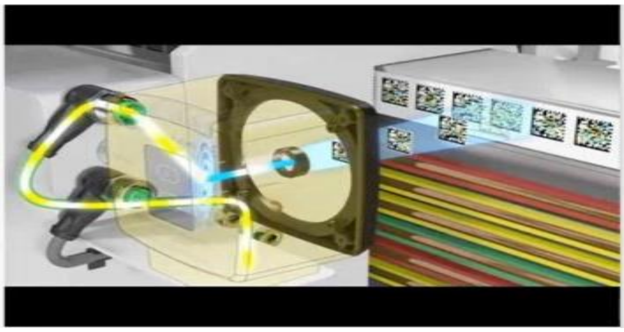
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the super master carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.





12.2.8 Carriers positioning System.

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



12.2.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	Loop CBS
Sorter Speed in m/s	Up to 2.3 m/s
Sorter Loop Length	~438 m
Sorter Actuation Technology	Electric
Chutes	
Outbound Chute	10
Staging Chute	40
Rejection Chute	2
Mixed Bag Chute	1
DBO Chute	1
Height of Sorter Upper Deck	~6500 mm
Height of Sorter Lower Deck	~4000 mm
Track Material to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch - mm	1175
No. of Carriers	373 Nos. per deck
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during Detailed Design
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Empty Carrier Detection System (Camera Based)	01 Nos per Zone
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Refer Layout

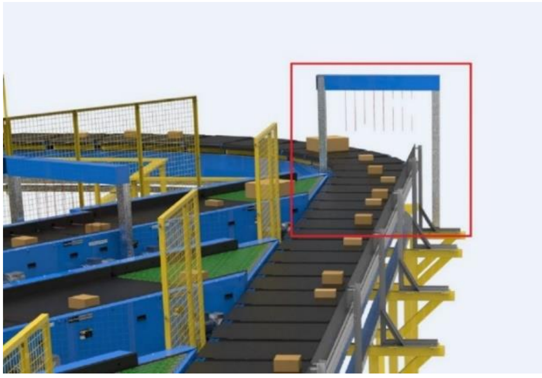
12.3 Scanning & Sensing on Main Sorter Loop

12.3.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.

12.3.2 Parcel Positioning System.

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).



12.3.3 Empty Carrier Detection System

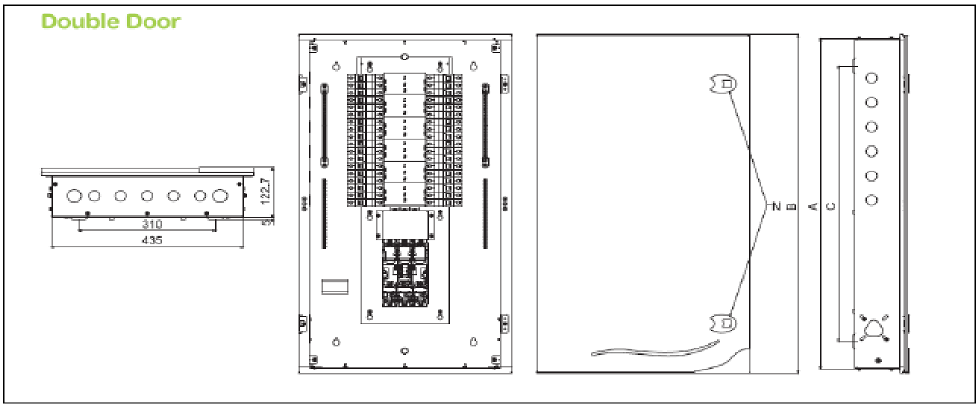
This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)

13. Electrical System

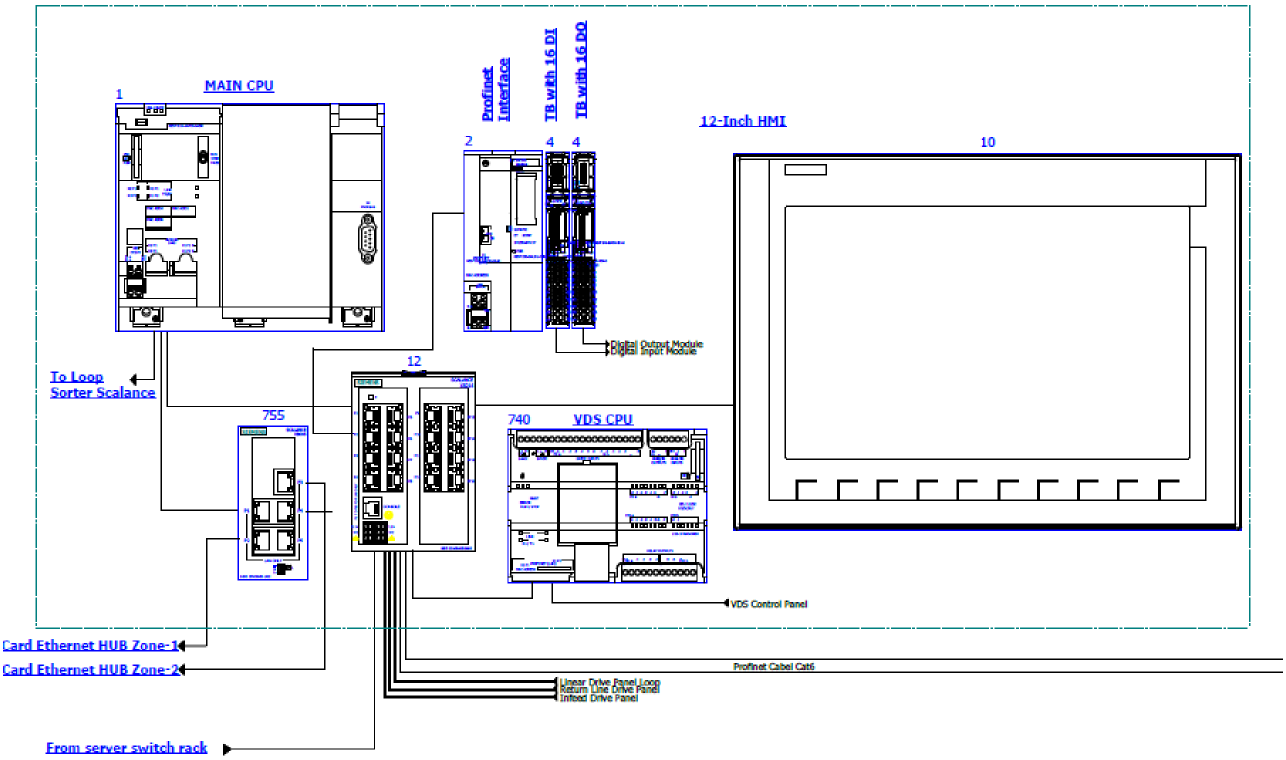
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

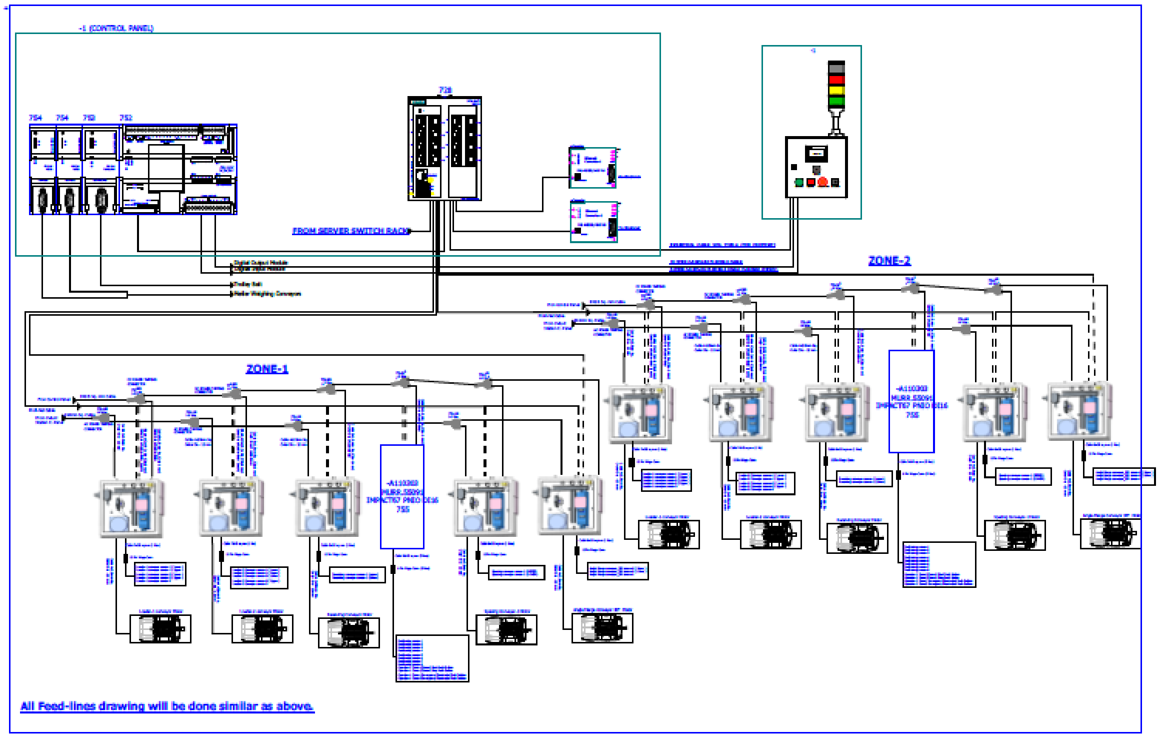
13.1.1 Reference Picture of Power Distribution Panel



13.1.2 Main Control Panel (Reference)



13.1.3 Induct Stations Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

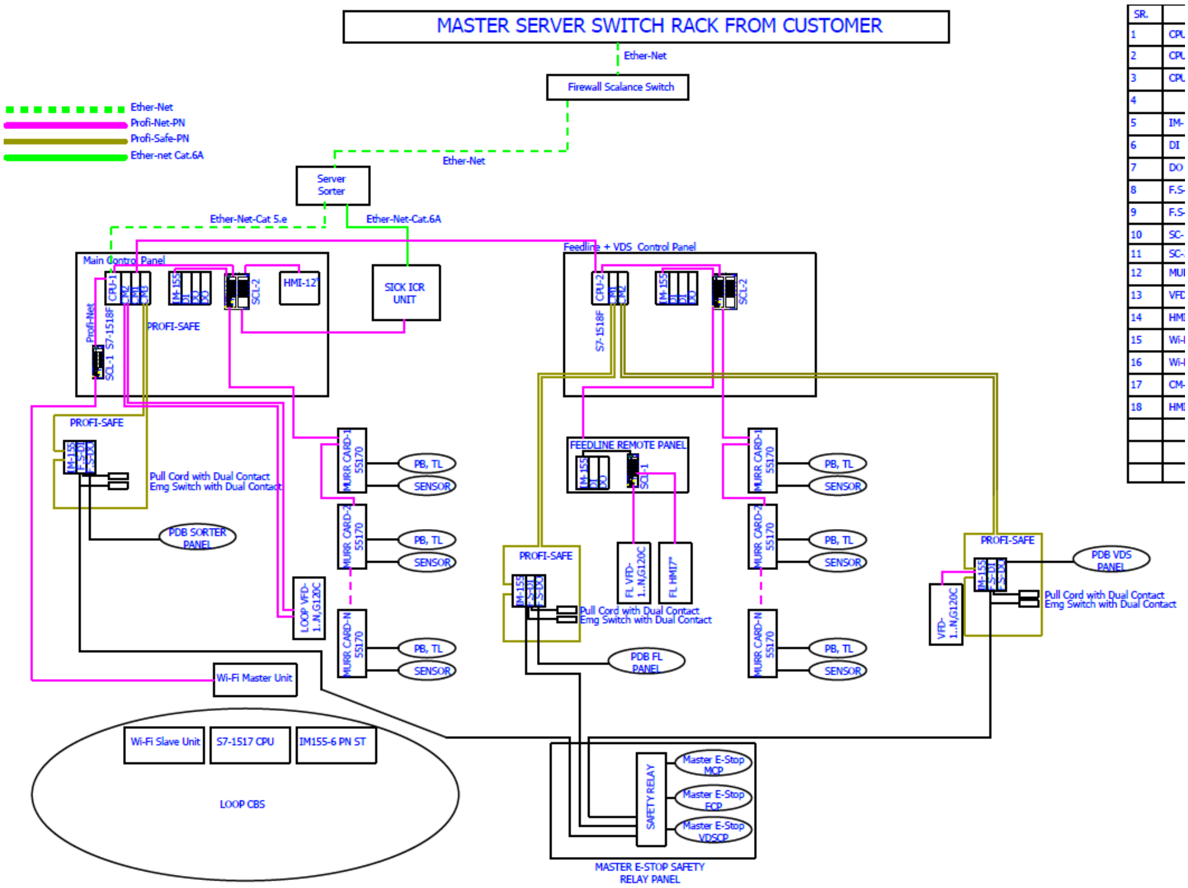
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into Zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

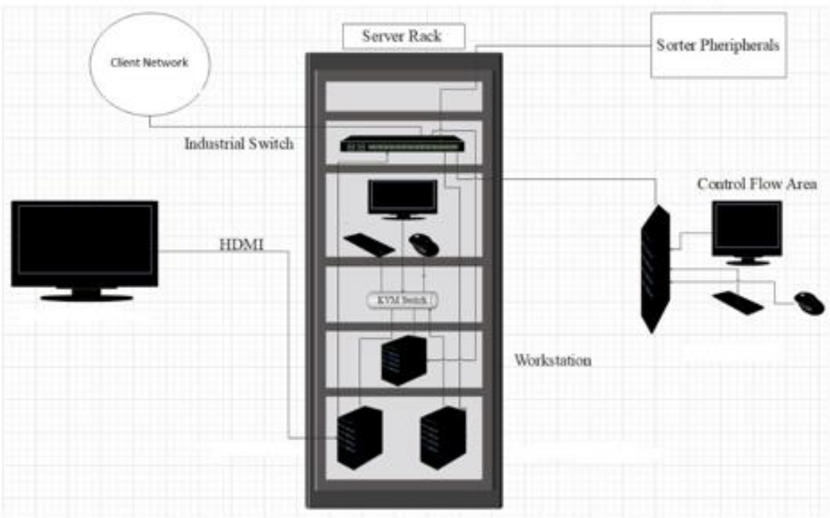
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

14. Controls Architecture (Ref.)



15. IT Architecture (Ref.)



16. Falcon's Visual Inspection System (SCADA)

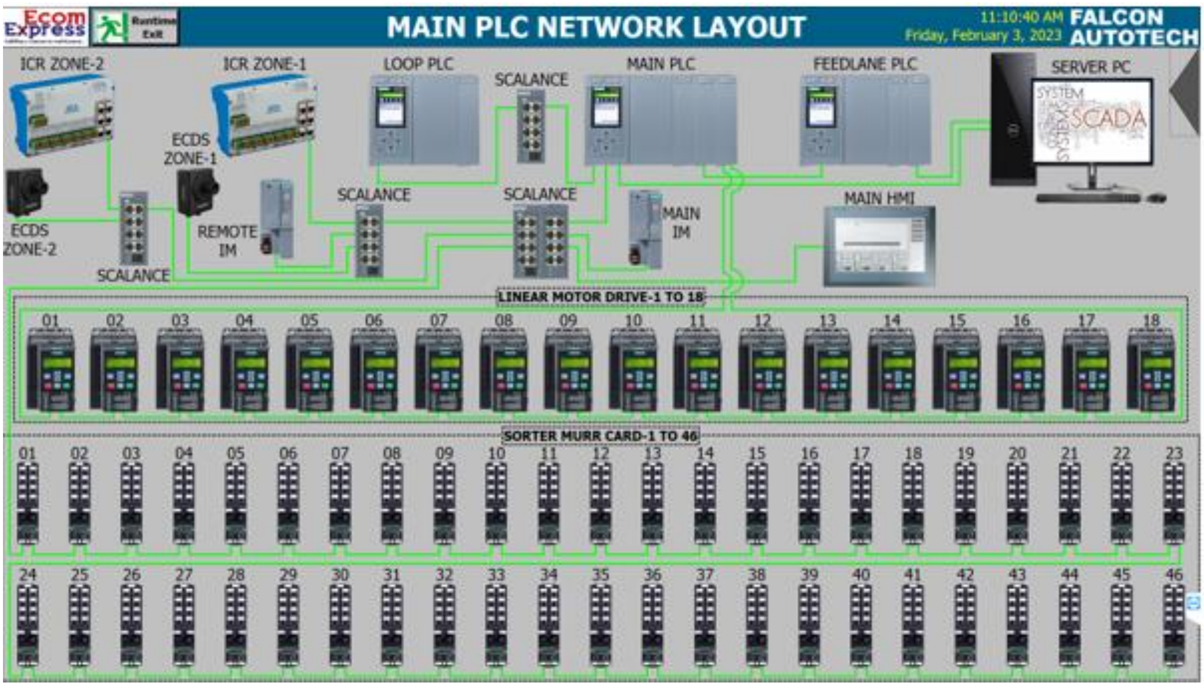
SCADA stands for Supervisory Control and Data Acquisition. It is a system of hardware and software components that allows for remote monitoring, control, and data acquisition of industrial processes or facilities

The Visualization system provided by FALCON (or SCADA) allows the monitoring and control of the different systems delivered for the St. Aunes Hub. This SCADA system receives from each monitored sub-system all information on their operating status in real time.
At the system monitoring level, the functions performed are:

- Field data acquisition.
- Animated visualization of equipment.
- Representation of the operating mode of the system (nominal, contingency, etc.).
- Alarm management.
- Alarm history management.
- Diagnostic help.
- Failure detection.
- Equipment control.
- Statistics on equipment operation.
- Historical Statistical Report.
- Recording and archiving.
- Safety operator interface.

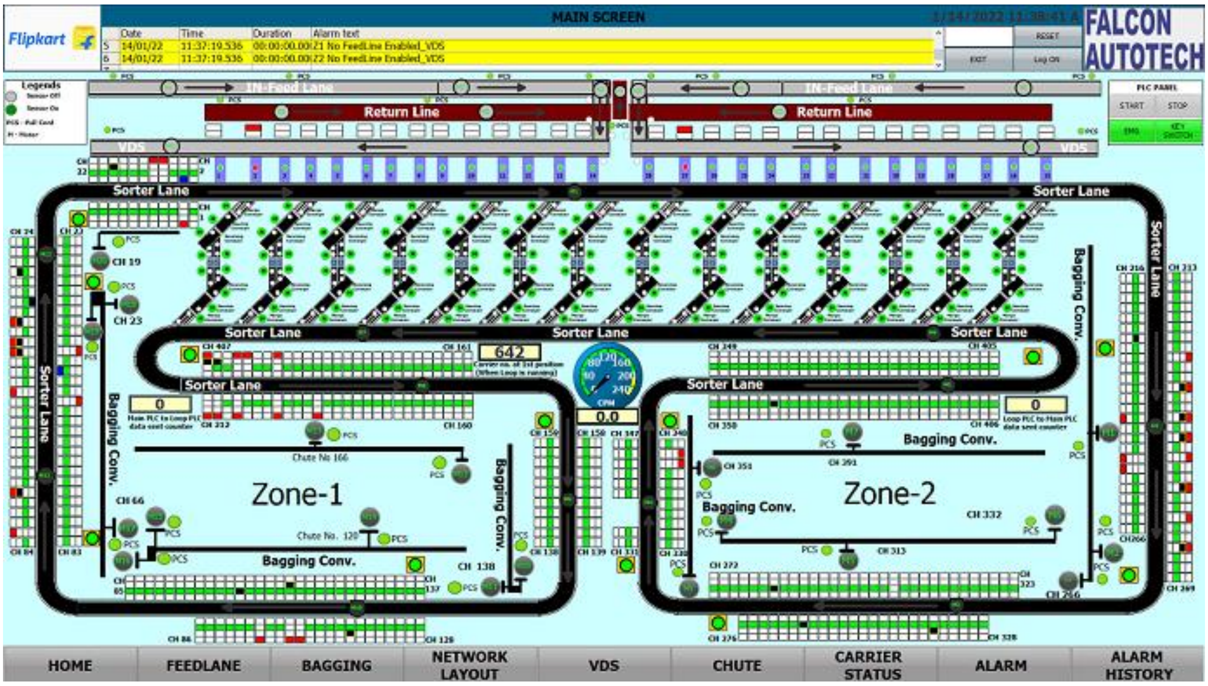
16.1 FIELD DATA ACQUISITION

The field data acquisition function is performed by the SCADA system connected to the sorters' PLCs. The communication with the PLCs is done using equipped CPU cards that are able to manage the communication with the PLC on the Industrial Ethernet network, without overloading the server.
The acquisition of signals from external systems is carried out via the PLC, using dry contacts.

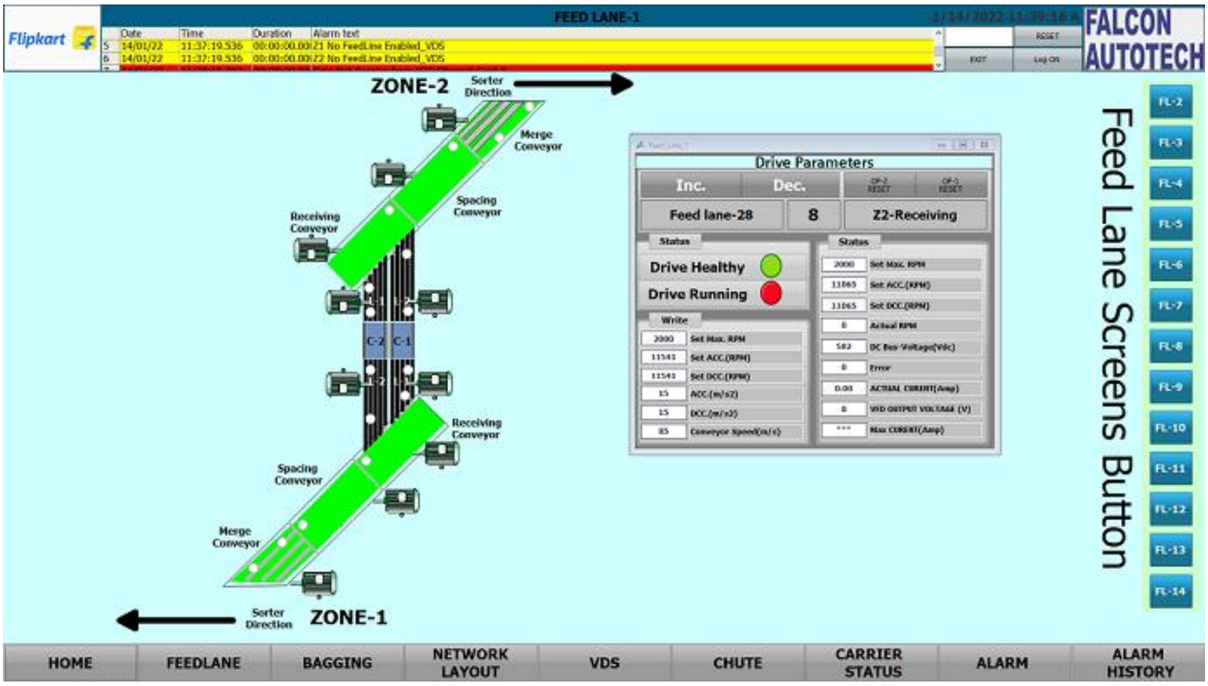


16.2 ANIMATED SYSTEM VISUALIZATION

The animated view represents the dynamic graphical user interface that allows real-time monitoring of the controlled systems and the execution of their control procedures.
The various states of the digital signals are displayed on the screen by graphic symbols. All views are web based with responsive capabilities when required so that various devices can be used to display status and information on PC. The types of information that can be displayed on each type of device will be discussed during the design phase of the project.



Example of a synoptic view



16.3 ALARM MANAGEMENT

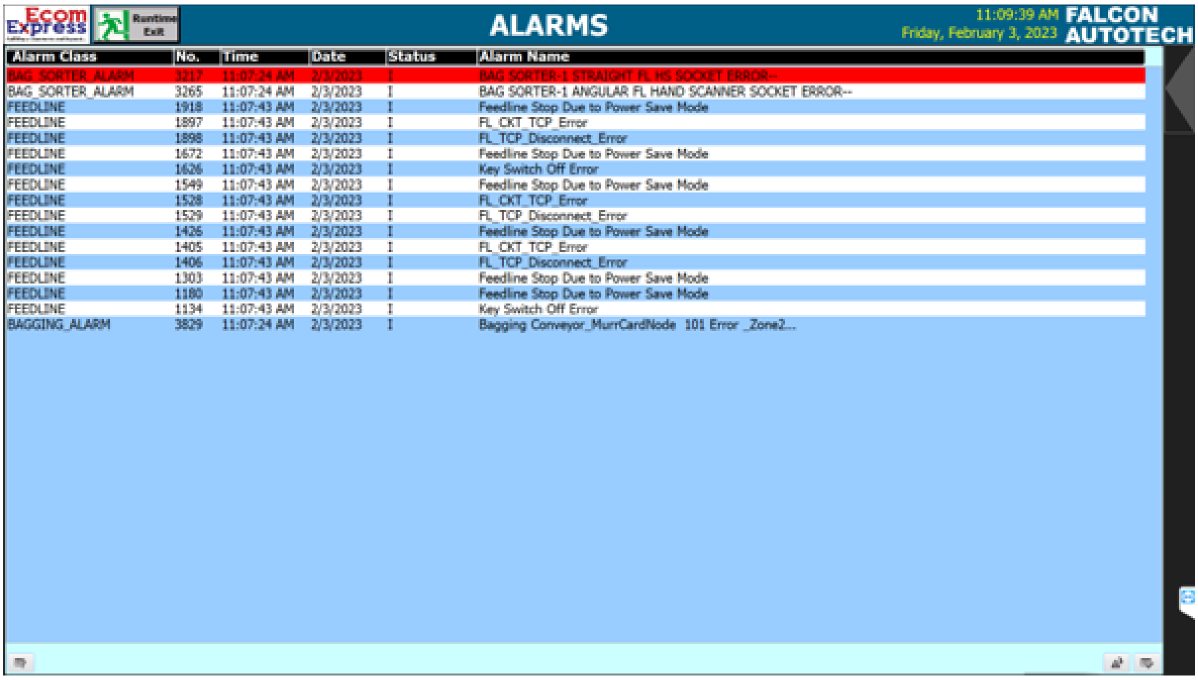
The alarm pages display a series of information to identify the nature of the alarm or event, the elements involved and the time.

The alarm management includes:

- The Alarm name.
- The date on which the above-mentioned alarm was activated.
- The moment the alarm goes off.
- The date on which the alarm is activated again.
- The moment the alarm is activated again.
- The sub-system that activated the alarm.
- The area where the alarm was triggered.
- The name of the beacon that activated the alarm.
- The alarm state.
- The alarm value.
- A complete alarm description.

The historic data of each alarm is stored in the SCADA database. This type of fault detected includes:

- ✓ Sensors.
- ✓ Drive Fault.
- ✓ Lack of monitored equipment.
- ✓ Etc.



ALARMS

11:09:39 AM Friday, February 3, 2023

Alarm Class	No.	Time	Date	Status	Alarm Name
BAG_SORTER_ALARM	3217	11:07:24 AM	2/3/2023	I	BAG SORTER-1 STRAIGHT FL HS SOCKET ERROR--
BAG_SORTER_ALARM	3265	11:07:24 AM	2/3/2023	I	BAG SORTER-1 ANGULAR FL HAND SCANNER SOCKET ERROR--
FEEDLINE	1918	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1897	11:07:43 AM	2/3/2023	I	FL_CKT_TCP_Error
FEEDLINE	1898	11:07:43 AM	2/3/2023	I	FL_TCP_Disconnect_Error
FEEDLINE	1672	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1626	11:07:43 AM	2/3/2023	I	Key Switch Off Error
FEEDLINE	1549	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1528	11:07:43 AM	2/3/2023	I	FL_CKT_TCP_Error
FEEDLINE	1529	11:07:43 AM	2/3/2023	I	FL_TCP_Disconnect_Error
FEEDLINE	1426	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1405	11:07:43 AM	2/3/2023	I	FL_CKT_TCP_Error
FEEDLINE	1406	11:07:43 AM	2/3/2023	I	FL_TCP_Disconnect_Error
FEEDLINE	1303	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1180	11:07:43 AM	2/3/2023	I	Feedline Stop Due to Power Save Mode
FEEDLINE	1134	11:07:43 AM	2/3/2023	I	Key Switch Off Error
BAGGING_ALARM	3829	11:07:24 AM	2/3/2023	I	Bagging Conveyor_MurrCardNode 101 Error _Zone2...

Sample Alarm History Screen

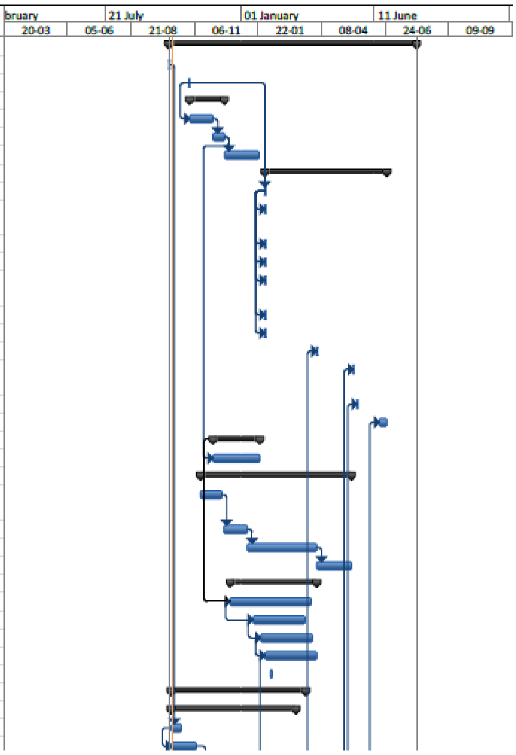
17. Key Components Make

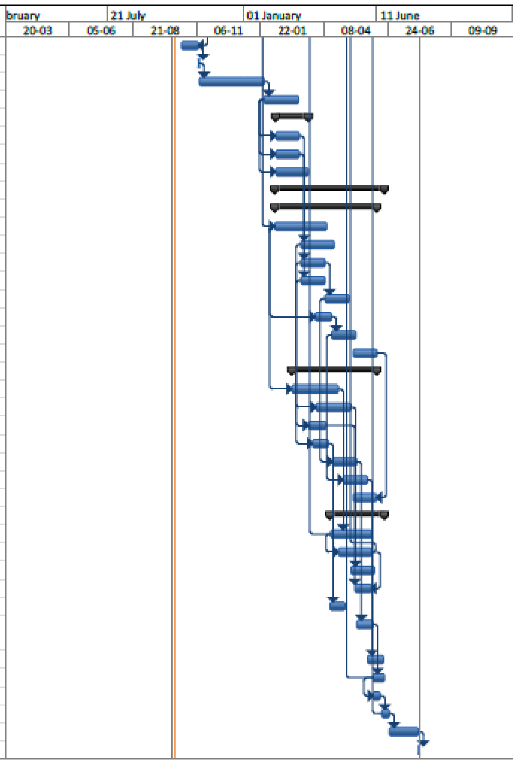
Items	Make
Belts	Forbo/ Derko
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Volume Scanners	SICK
Encoders	SICK/Falcon
Sensors	Sick/Leuze/ P&F
PLC	Siemens's
Control Panels	Rittal/ BCH
Weighing Scales	Bizerba/Equivalent
VFDs	Siemens/Lenze
Cables	LAPP Equivalent
Switch Gear	Schenider/Equivalent
Bearings	NTN/SKF/Equivalent
Data Transmission System	Siemens
Power Transmission Systems	Vahle
HMIs	Siemens

18. Program Organisation

18.1 Program/ Project Schedule

D	Task Name	Duration	Start	Finish	Predecessors
1	**Delhivery Bangalore Bag Sorter**	**214 days**	**Fri 06-10-23**	**Wed 31-07-24**	
2	Solution Submission	1 day	Fri 06-10-23	Fri 06-10-23	
3	**Kick Off Meeting**	**1 day**	**Mon 30-10-23**	**Mon 30-10-23**	
4	**DAP Activities**	**30 days**	**Tue 31-10-23**	**Mon 11-12-23**	
5	DAP SUBMISSION	20 days	Tue 31-10-23	Mon 27-11-23	3SS+1 day
6	DAP APPROVAL	10 days	Tue 28-11-23	Mon 11-12-23	5
7	**IT-BRD Submission**	**30 days**	**Tue 12-12-23**	**Mon 22-01-24**	**6**
8	**Requirements from Delhivery**	**106 days**	**Tue 30-01-24**	**Tue 25-06-24**	
9	Handover of Laydown Area & Safe Room	1 day	Tue 30-01-24	Tue 30-01-24	3FS+60 days
10	Availability of Basic Amenities (washrooms, sitting infra, drinking water)	1 day	Tue 30-01-24	Tue 30-01-24	9SS
11	Availability of Working lights on Floor	1 day	Tue 30-01-24	Tue 30-01-24	9SS
12	Availability of Ramp for MHE & Hydra movement	1 day	Tue 30-01-24	Tue 30-01-24	9SS
13	Availability of access roads & Docks with Leveller (or Dock Plates)	1 day	Tue 30-01-24	Tue 30-01-24	9SS
14	Availability of CCTV & Security Guard	1 day	Tue 30-01-24	Tue 30-01-24	9SS
15	Availability of Power (RAW & Earthing)	1 day	Tue 30-01-24	Tue 30-01-24	9SS
16	Availability of Power (UPS & Earthing)	1 day	Tue 02-04-24	Tue 02-04-24	64SS-10 days
17	Availabity of IT Infra (Server Room + Ethernet Points)	1 day	Thu 16-05-24	Thu 16-05-24	71SS-15 days
18	Availability of Dummy Shipments	1 day	Tue 21-05-24	Tue 21-05-24	65FS-10 days
19	Availability of Live Shipments	7 days	Mon 17-06-24	Tue 25-06-24	73SS
20	**Detailed Engineering**	**41 days**	**Tue 28-11-23**	**Tue 23-01-24**	
21	Detailed Engineering	41 days	Tue 28-11-23	Tue 23-01-24	6FS-10 days
22	**Spiral Chutes**	**131 days**	**Mon 13-11-23**	**Mon 13-05-24**	
23	Collecting Quotations & Drawings from Vendor	20 days	Mon 13-11-23	Fri 08-12-23	
24	Negotiation & PO placement for chute	21 days	Mon 11-12-23	Mon 08-01-24	23
25	Production of Spiral Chutes	60 days	Tue 09-01-24	Mon 01-04-24	24
26	Transit Time for Spiral Chutes	30 days	Tue 02-04-24	Mon 13-05-24	25
27	**Manufacturing, procurement and dispatch**	**75 days**	**Tue 19-12-23**	**Mon 01-04-24**	
28	Manufacturing and procurement	70 days	Tue 19-12-23	Mon 25-03-24	20SS+15 days
29	Assembly	45 days	Tue 16-01-24	Mon 18-03-24	28SS+20 days
30	Packing	45 days	Thu 25-01-24	Wed 27-03-24	29SS+7 days
31	Dispatch	45 days	Tue 30-01-24	Mon 01-04-24	30SS+3 days
32	**Layout Marking**	**2 days**	**Tue 06-02-24**	**Wed 07-02-24**	
33	**Mezzanine**	**117 days**	**Mon 09-10-23**	**Tue 19-03-24**	
34	**Design, Planning & Procurement**	**109 days**	**Mon 09-10-23**	**Thu 07-03-24**	
35	Vendor Finalisation	10 days	Mon 09-10-23	Fri 20-10-23	2
36	Mezzanine Design	22 days	Mon 09-10-23	Tue 07-11-23	35SS
37	Price negotiation	15 days	Wed 18-10-23	Tue 07-11-23	36FF
38	PO placement	1 day	Wed 08-11-23	Wed 08-11-23	37
39	Mezzanine Production	56 days	Thu 09-11-23	Thu 25-01-24	38
40	Mezzanine Material Supply	30 days	Fri 26-01-24	Thu 07-03-24	39
41	**Installation**	**28 days**	**Fri 09-02-24**	**Tue 19-03-24**	
42	Inbound & Highway	21 days	Fri 09-02-24	Fri 08-03-24	40SS+10 days
43	Mezzanine Feedline Zone-1	21 days	Fri 09-02-24	Fri 08-03-24	40SS+10 days
44	Mezzanine Feedline Zone-2	28 days	Fri 09-02-24	Tue 19-03-24	40SS+10 days
45	**Installation & Commissioning**	**95 days**	**Thu 08-02-24**	**Wed 19-06-24**	
46	**Installation Mechanical**	**88 days**	**Thu 08-02-24**	**Mon 10-06-24**	
47	CBS	45 days	Thu 08-02-24	Wed 10-04-24	31SS+7 days
48	IB & Highway + Loop VDS	30 days	Mon 11-03-24	Fri 19-04-24	42
49	Manual Feedline Zone-1	21 days	Mon 11-03-24	Mon 08-04-24	43
50	Auto Feedline Upper Deck	21 days	Mon 11-03-24	Mon 08-04-24	43
51	Manual Feedline Zone-2	21 days	Tue 09-04-24	Tue 07-05-24	49
52	Outbound Conveyors Legs	14 days	Thu 28-03-24	Tue 16-04-24	47SS+35 days
53	Outbound Conveyors	21 days	Wed 17-04-24	Wed 15-05-24	52
54	Finishing Activities	21 days	Mon 13-05-24	Mon 10-06-24	
55	**Installation Electrical**	**73 days**	**Thu 29-02-24**	**Mon 10-06-24**	
56	CBS	40 days	Thu 29-02-24	Wed 24-04-24	47SS+15 days
57	IB & Highway + Loop VDS	30 days	Fri 29-03-24	Thu 09-05-24	48SS+14 days
58	Manual Feedline Zone-1	15 days	Wed 20-03-24	Tue 09-04-24	49SS+7 days
59	Auto Feedline Upper Deck	15 days	Mon 25-03-24	Fri 12-04-24	50SS+10 days
60	Manual Feedline Zone-2	21 days	Thu 18-04-24	Thu 16-05-24	51SS+7 days
61	Outbound Conveyors	21 days	Wed 01-05-24	Wed 29-05-24	53SS+10 days
62	Finishing Activities	21 days	Mon 13-05-24	Mon 10-06-24	54FF
63	**Commissioning**	**48 days**	**Mon 15-04-24**	**Wed 19-06-24**	
64	I/O testing	35 days	Tue 16-04-24	Mon 03-06-24	56FS-7 days
65	Sorter and Feedline programming and Testing	28 days	Thu 25-04-24	Mon 03-06-24	64SS+7 days
66	IB & Highway + Loop VDS	21 days	Fri 10-05-24	Fri 07-06-24	57
67	Manual Feedline Zone-1 synchronisation	14 days	Wed 15-05-24	Mon 03-06-24	58,65FF
68	Auto Feedline Upper Deck synchronisation	14 days	Mon 15-04-24	Thu 02-05-24	59
69	Manual Feedline Zone-2 synchronisation	14 days	Fri 17-05-24	Wed 05-06-24	60
70	Outbound Conveyors	14 days	Thu 30-05-24	Tue 18-06-24	61
71	IT + PLC communication testing	10 days	Thu 06-06-24	Wed 19-06-24	69
72	**Trial with Dummy shipments**	**7 days**	**Thu 06-06-24**	**Fri 14-06-24**	**71SS**
73	**Pre-UAT**	**7 days**	**Mon 17-06-24**	**Tue 25-06-24**	**72**
74	**UAT & Operations Scale-Up**	**25 days**	**Wed 26-06-24**	**Tue 30-07-24**	**73**
75	**GO-LIVE**	**1 day**	**Wed 31-07-24**	**Wed 31-07-24**	**74**





**Above plan is tentative, detailed plan will be shared post order confirmation.*

18.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with DELHIVERY
3. Weekly/Fortnightly Meeting DELHIVERY to share the Project Status.
4. Scheduling of the resource management.
5. Management of risks and opportunities.
6. Management of the requirements.
7. Management of the list of anomalies or reservations.

19. DELHIVERY'S Responsibility

19.1 DELHIVERY Responsibilities During the Assembly and Commissioning Phase

- Provision of the site complex and office area facilities.
- The possibility of authorizing access to the site and the execution of the installation work up to 7 days a week and 24 hours a day if deemed necessary and if requested by FALCON.
- Free provision, during the installation phase, of the power supply necessary for the installation activities (estimated at 20 kW).
- Provision, during the commissioning phase, of the power supply necessary for the operation of the parcelsorting system free of charge at the date of FALCON need.
- Provision of the IT system functionality in accordance with the specification at the date of FALCON need.
- The customer is responsible for a safe working environment.
- The customer makes arrangements for the working area('s) to be protected against direct weather influences.
- The customer provides adequate lighting, heating, and ventilation to create a normal working environment.
- The cost of temporary storage that may be required (other than in the immediate vicinity of the installationsite) is not included in the scope of delivery of this quotation.

This also applies to temporary storage that may be required because materials are ready for delivery (in accordance with the schedule) but cannot be delivered due to hold-ups on the customer's side.

- The customer is responsible for the demarcation of aisles and danger zones with floor paint.

19.2 Responsibilities of DELHIVERY During the Tests

- Provision of the test loads and barcode labels required for the tests.
- Provision of personnel required for test activities (loading and unloading operations).
- Provision of the necessary information to sort the parcels correctly.
- Verify with FALCON the quality and conformity of the test loads (labels, cartons).
- Will need to provide the necessary staff to collect information on the tests and to verify and confirm testresults with FALCON.

19.3 DELHIVERY Responsibilities During the Training

- Free from their usual work, the employees participate in the training for the duration of the training.
- Provision of a list of participants for each available training course 3 days before the start of the course.
- Provision of a classroom equipped with a whiteboard, video projector, projection screen, and enough spacefor desks or tables and chairs for the trainer and trained staff.
- check the prerequisites of the people who have to follow the training, e.g. the qualifications of the technicalstaff.
- The invitation of staff to attend the training courses will be at the expense of DELHIVERY.

20. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (**Starts from the date of beneficiary use)** and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
    - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
    - Negligence or abnormal use of equipment.
    - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
    - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

21.Commercial Terms

21.1 Price Sheet

Phase 1:

S. No	Component	Qty per set	Set	UOM	Total(INR)
1	Loop Cross Belt Sorter with VMS, ICR(Upper Deck)	1	1	Set	₹ 13,16,72,730
2	Loop Cross Belt Sorter Only Track(Lower Deck)	1	1	Set	₹ 1,74,46,672
3	Conveyors & Swedi	1	1	Set	₹ 9,83,81,980
4	Chutes & Its Accessories	1	1	Set	₹ 4,55,06,950
5	Telescopic Belt Conveyor System (7+14)	1	23	Nos	₹ 3,31,20,000
6	Software Packages & Tool Kit	1	1	Set	₹ 11,79,800
7	Installation & Commissioning	1	1	Set	₹ 97,83,850
8	Packaging & Forwarding	1	1	Set	₹ 32,62,081
9	Project Management and Engineering Charges	1	1	Set	₹ 17,01,770
Total					**₹ 34,20,55,834**
BCA Discount (5 %)					**₹ 1,71,02,792**
Final					**₹ 32,49,53,043**

Phase 2:

S. No	Component	Qty per set	Set	UOM	Total (INR)
1	Loop Cross Belt Sorter (Lower Deck)	1	1	Set	₹ 9,38,88,629
2	Conveyors & Swedi	1	1	Set	₹ 7,13,95,937
3	Chutes & Its Accessories	1	1	Set	₹ 6,30,630
4	Telescopic Belt Conveyor System (7+14)	1	12	Nos	₹ 1,72,80,000
5	Software Packages & Tool Kit	1	1	Set	₹ 3,00,000
6	Installation & Commissioning	1	1	Set	₹ 64,11,832
7	Packaging & Forwarding	1	1	Set	₹ 18,31,952
8	Project Management and Engineering Charges	1	1	Set	₹ 19,17,390
Total					**₹ 19,36,56,369**
BCA Discount (5 %)					**₹ 96,82,818**
Final					**₹ 18,39,73,551**

The Total Price is to be understood.

- Freight- Excluded
- Taxes: Extra as Applicable
- Including Packing and Forwarding
- Installation and Commissioning: Included
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area are in customer scope

21.2 Payment Terms

S.No	Percentage	Stage
1	40%	Along with Purchase Order within 7 days of contract signing
2	50%	Against Delivery of material at site with 100% Taxes on Pro Data Basis
3	10%	Against Installation & Commissioning

**All invoices (if due as per payment terms) should be cleared within 30 days from the date of receipt of undisputed invoice by the buyer*

22. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Construction Power**
- **Mezannine**
- **Steel Works- If not specified**
- **Collection Trolleys below chutes.**
- **PC for SCADA.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets.
- UPS for Controls and Drives
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- **Simulation and 3D animation of the sorter system**
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.